

BBE SCHOOL

Introduction to Business and Economics

Study material for the Economics Full Course
Chapters 2 to 6

An original course textbook prepared for this platform. It covers the same syllabus progression as the WU BBE entrance pathway in a calm, structured form: chapters, definitions, worked examples, tables and figures.

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Chapter 2

Basic economic concepts

Before one can study how a business is organised and managed, it is necessary to understand the economic environment of which every business forms a part. Basic economic concepts shape the decisions of firms and of individuals: what to offer for sale, what to purchase, how to manage limited resources, and how to finance expenditure.

This chapter develops that shared foundation. It begins with participation in the economy and the condition of scarcity that forces every household, firm and government to choose. It then presents economics as the study of such decisions, and shows how exchange creates a circular flow of goods, services and money, together with division of labour. Different economic systems allocate decision rights between private agents and the state in different ways. Markets coordinate buyers and sellers through supply and demand, and competition among suppliers shapes the setting in which those markets operate. The aim is a clear, connected understanding of these ideas rather than a list of slogans.

2.1 Being part of the economy

An economy is the system in which people and organisations produce, distribute and exchange goods and services in order to satisfy needs and wants. Running a business is one way of taking part in that system, but it is by no means the only way. Long before anyone founds a firm, people belong to private households that purchase food, housing, transport, medical care, education and leisure. Households also supply labour, skills and savings. In that dual role-demanders of goods and services and suppliers of resources-almost everyone is already an active participant.

Businesses provide the goods and services that households and other organisations cannot produce efficiently alone. A bakery supplies bread; a logistics firm supplies delivery; a software studio supplies applications and maintenance. Businesses themselves have needs: they require materials, energy, premises, finance and a workforce. One firm's output frequently becomes another firm's input. Manufacturing depends on mining or farming; retail and wholesale depend on manufacturing; services wrap around almost every stage. Chains of interdependence are therefore the normal structure of economic life, not an exception.

Business. A business is an entity that offers goods and/or services to customers. It produces primarily for customers rather than only for the owners' private household needs, and usually charges a price so that something of value is received in return.

Economy. The economy is the overall system of production, exchange and consumption through which households, businesses and public authorities interact to meet needs and wants.

Exchange is not confined to formal firms. Private persons sell a used bicycle, rent a room, or swap garden produce for a bottle of wine. Those private exchanges still belong to economic life because scarcity, valuation and mutual benefit are present. Exchanging goods and services against money or another means of payment is a central characteristic of the modern economy.

Exchange would be much easier if everything people want were available in abundance. That is not the case. Resources are scarce and must be managed. For that reason everyone must economise: no household, firm or government can wholly opt out of economic decisions. It is a common mistake to treat "the economy" as something that happens only on stock markets or only among entrepreneurs. Employees, students who buy services, pensioners who consume goods, and taxpayers who help finance public goods are all participants.

Worked example. A student works evenings in a café and spends wages on rent, food and a language course. She already takes part in the economy as a supplier of labour and as a consumer. If she later opens a tutoring service for secondary-school pupils, she remains a household member who buys transport

and phones, but she additionally becomes an entrepreneur supplying education services. Both roles belong to the same economic system; only the mix of activities changes.

2.2 Scarcity of resources and opportunity cost

Both households and businesses have only limited resources with which to pursue their goals. A firm has a finite number of machines, a limited stock of materials and constrained financial means. A household has a finite income that can be spent on goods and services, with perhaps a part set aside for saving. Governments, too, face limits of tax revenue and borrowing capacity in any given period. Scarcity means that resources available to meet goals are limited relative to wants. Scarcity forces allocation: deciding which uses win and which are postponed or rejected.

Scarcity. Scarcity is the basic condition that resources are limited while needs and wants are effectively unlimited, so that choices about allocation become necessary.

It is a basic economic problem for households, businesses and the government to decide how to use limited resources and to allocate them among different options. A graduate may found a business or take paid employment. A firm may invest in a delivery vehicle or keep funds in a safer interest-bearing account. A parliament may expand one public programme only by reducing another, raising taxes, or accepting more debt. In every such case the decision is not free of consequence: choosing one path means giving up another.

Opportunity cost. Opportunity cost is the benefit of the next-best alternative that is lost or given up when a choice is made. It is measured not only in money but also in time, comfort, risk or other valued outcomes of the forgone option.

Opportunity cost makes trade-offs visible. Spending a Saturday revising accounting rather than working an extra shift has an opportunity cost in wages not earned, and possibly in leisure. Founding a firm instead of taking a salaried job means giving up that salary and its associated security-even if the new venture's accounts show no wage expense for the founder. Opportunity cost is often hardest to see when no cash changes hands: studying may look "free" of tuition fees at one institution, yet the next-best paid job still has a value that is given up.

Two clarifications matter. First, opportunity cost is the next-best alternative-the best option among those left behind-not the sum of every rejected dream. Second, scarcity is not the same as poverty. Even wealthy households and rich countries face scarcity because wants expand and resources remain finite for any given decision. It is a common mistake to claim that there is "no cost" whenever no money was paid. If a realistic alternative was deliberately given up, opportunity cost is present.

Worked example. *Two graduates can each earn €38,000 a year in employment. If they start a repair café instead, the opportunity cost of self-employment is at least that salary each-plus any benefits they would have received-because that package is the next-best realistic alternative. If the business later earns more than the combined forgone salaries after proper cost allowance, the choice can still be rational; the point is that zero reported wage for the founders does not mean zero cost of their time.*

2.3 Economics is the study of economic decisions

Economics is the study of how individuals (as members of private households), businesses and societies decide how to satisfy needs and wants with limited resources. It comprises a number of scientific fields and branches. Two that appear constantly in introductory work are microeconomics and macroeconomics. The discipline is both descriptive-explaining observed behaviour-and analytical-building models that isolate key relationships and generate predictions under stated assumptions.

Economics. Economics is the study of how agents make decisions to allocate scarce resources in order to satisfy needs and wants, and of how those decisions interact in markets and across the economy as a whole.

Microeconomics focuses on the behaviour and decisions of individual households and businesses and on how they interact in particular markets. A typical microeconomic question asks how demand for electric cars would change if buyers received a purchase bonus. Macroeconomics looks at the larger picture: economic growth, unemployment, interest rates, the overall price level and inflation, among other aggregate phenomena. The distinction is individual or market versus economy-wide, not "small firms" versus "large firms".

Microeconomics. Microeconomics focuses on individual households and businesses, their decisions, and how they interact in particular markets.

Macroeconomics. Macroeconomics studies the economy in aggregate: growth, unemployment, interest rates, the overall price level and inflation, among other economy-wide phenomena.

The two lenses are complementary. A wage rise in one firm is a microeconomic issue; rising unemployment nationwide is macroeconomic. Yet firm-level hiring accumulates into national employment figures, and national interest-rate policy feeds back into household borrowing and firm investment. As a science, economics uses theories and simplifying assumptions. Models deliberately omit detail so that cause and effect become visible. Predictions are conditional: if incomes rise and tastes stay constant, demand for normal goods tends to rise. Understanding the "if" is as important as remembering the conclusion.

Economic decisions appear everywhere: whether to hire or automate, rent or buy, specialise or diversify, tax or borrow, consume now or save. Business studies then ask how organisations design products, finance assets and manage people inside that environment. Chapter 2 therefore supplies the common vocabulary before later chapters turn to ownership, marketing and accounting.

Worked example. Question A asks how a €2,000 purchase bonus for electric bikes would affect demand for e-bikes in one city. That is microeconomics-buyers and sellers in one product market. Question B asks how inflation and unemployment evolved in the euro area last year. That is macroeconomics-aggregate indicators for a large economic area.

2.4 Exchanging goods and services creates a circular flow and division of labour

Exchange is the bridge that turns specialisation into a workable system. Households typically supply labour and receive wages; businesses supply goods and services and receive revenue. Those flows of real resources and money move in opposite directions around the same relationships. Households provide labour to firms and receive wages; firms provide goods and services to households and receive payment. This continuous movement of resources, goods and services in one direction and monetary payments in the other is the circular flow.

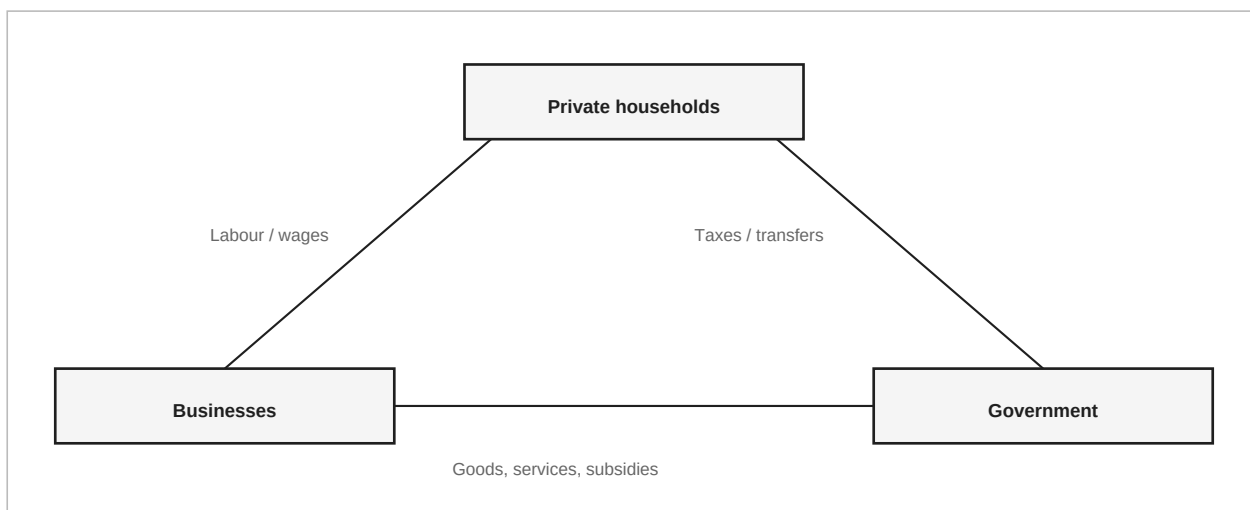


Figure 1. The circular flow of goods, services and money within the economy - households, businesses and government linked by labour, wages, products, payments, taxes and public services.

Circular flow. The circular flow describes the continuous movement of resources, goods and services in one direction and monetary payments in the opposite direction between households, businesses and government.

Public authorities complete the picture. Governments levy taxes from households and firms and use the proceeds to provide public goods, transfer payments and subsidies-roads, defence, policing, and often large parts of education and healthcare. Some goods create free-rider problems: people can benefit without paying voluntarily, so private firms alone would underprovide them. Taxation finances those services when markets would not deliver adequate supply on their own.

Money makes this pattern workable. As a widely accepted means of payment it allows flexible trade without barter. Without money, exchange requires a double coincidence of wants: each party must want what the other offers at the same time. Money breaks that coincidence. It also serves as a unit of account, expressing and comparing values through prices, and as a store of value, carrying purchasing power into the future when its value remains reasonably stable.

Money (three functions). Money is a widely accepted means of payment. It serves as (1) a medium of exchange, allowing flexible trade without barter; (2) a unit of account, expressing and comparing values; and (3) a store of value, carrying purchasing power into the future when its value remains reasonably stable.

Money fulfils these functions best when its purchasing power is fairly stable. A general rise in the prices of goods and services-inflation-reduces that purchasing power. Price indexes measure how strong the rise is. Mild inflation can be tolerated; the European Central Bank regards an inflation rate of slightly below 2% per year as generally beneficial. High inflation erodes trust: people rush to spend before prices rise further. It is a common mistake to confuse a price rise for one product with inflation; inflation is a general rise across many goods and services.

Inflation. Inflation is a general rise in the prices of goods and services, which reduces the purchasing power of money. Price indexes measure the extent of that general rise.

Division of labour and specialisation follow from exchange. If every household tried to produce all it needed, people would waste time on tasks for which they have weak skills. Exchange lets agents concentrate on comparative strengths. Specialisation appears within households (who shops, who cooks); within firms as departments such as procurement, production, sales, marketing, finance and human resources; between firms at the same production level or along a chain (timber -> boards -> furniture -> retail); and internationally, where climate, natural resources, know-how, labour costs and legal rules differ

across countries.

Division of labour raises productivity but carries risks. Highly specialised work can become monotonous. Workers may lose flexibility if narrow skills fall out of demand. A firm that masters one niche can be vulnerable if demand for that niche collapses. Specialisation is therefore powerful, not free of downside. Money's three functions facilitate exchange; they are not themselves a guarantee of wealth. Wealth creation comes from production and exchange.

Worked example. *A bicycle repair shop specialises in assembly and aftercare; a separate supplier makes frames; a courier firm delivers parts. Money lets each focus: the shop need not barter repaired bikes for tubing. If inflation jumps sharply, the shop's cash balances buy fewer spare parts over a few weeks, showing that the store-of-value function weakens when the general price level rises quickly.*

2.5 Different economic systems

An economic system answers who decides what is produced, how it is produced, and who receives the output. The two classic poles are the market economy and the planned economy. In market economies, individuals and businesses are more or less allowed to make many of their own economic decisions about production, consumption and prices within a legal framework. Markets coordinate allocation through supply and demand. In planned (or centrally planned) systems, government mainly or partly decides which goods and services are produced, at which prices, and controls key resources and means of production. Job choice and product choice for citizens are more limited than in market systems.

Market economy. In a market economy, individuals and businesses make many of their own economic decisions about production, consumption and prices, within a legal framework. Markets coordinate allocation through supply and demand.

Planned economy. In a planned (or centrally planned) system, government mainly or partly decides which goods and services are produced, at which prices, and controls key resources and means of production.

Most real countries sit somewhere between the pure textbook ends of the spectrum. Even within market-oriented countries, roles for the state differ. A free market economy keeps government intervention comparatively light: rules of property, contract and competition exist, but officials do not routinely redirect production or redistribute extensively. A social market economy (or eco-social market economy) keeps markets as the main coordinating device while giving the state a stronger role-supporting the vulnerable, correcting market failures, and protecting the environment, among other tasks.

Feature	Free market economy	Social market economy	Planned economy
Who mainly decides what is produced?	Private households and firms through markets	Private agents through markets; state sets social and environmental frameworks	Government (mainly or partly) through plans and quotas
How are prices determined?	Mostly by supply and demand	Mostly by supply and demand, with selected regulation and taxes	Often set or tightly controlled by the state
Role of government	Legal framework; comparatively light economic intervention	Legal framework plus stronger social protection, redistribution and environmental policy	Dominant control of production, resources and often employment
Ownership of means of production	Predominantly private	Predominantly private, with public enterprises in selected areas	Predominantly public or tightly state-controlled
Consumer and job choice	Wide	Wide, within legal and social rules	More limited

Table 1. Comparison of free market, social market and planned economic systems

Free market economy. A market system in which government mainly provides the legal framework and influences the economy relatively little beyond that framework.

Social market economy. A market system combined with a more active government role in social protection, environmental and related policy goals, while market exchange remains central to allocation.

Over recent decades, many formerly planned economies adopted market principles-across parts of Central and Eastern Europe, China and successor states of the Soviet Union, for example. Transformation is rarely overnight: legal institutions, property rights, competition culture and safety nets must catch up with price liberalisation. A social market economy is still a market economy, not a planned economy: extra social and environmental intervention does not of itself amount to central planning of all production. Likewise, the mere fact that government provides schools or roads does not make a system planned; public goods appear in market economies too.

***Worked example.** Country A lets private firms choose products and set most prices; consumers choose freely among suppliers; government mainly enforces contracts and competition law. That leans towards a free market pattern. Country B also uses private ownership and prices, but finances universal healthcare, generous unemployment support and tight environmental standards through taxes. That is closer to a social market pattern. Country C issues production quotas for steel and bread, sets official prices, and assigns workers to plants: that matches planned-economy features.*

2.6 Supply and demand: households, businesses and the government meet in the market

In a market economy, goods and services are offered and sold in markets. Buyers and sellers communicate the conditions of exchange and thus form a market. A market can be an actual place-a flower market in a city, a flea market in a village-or a virtual place such as an internet platform. Buying and selling take place in shops, by phone and online. Markets differ by what is traded: consumer goods, labour, housing, money, capital and commodities each have their own arenas. Households, businesses and government appear on both the demand and the supply side depending on the market considered.

Market. A market is the interaction of buyers and sellers who communicate and agree, implicitly or explicitly, on the terms of exchanging a good or service-whether face-to-face or through intermediaries and platforms.

Supply of a good or service is the quantity available for purchase. That quantity depends on production capacity and resources and, crucially, on the price that can be charged. All other things held constant-ceteris paribus-a higher price leads to a higher quantity supplied for most goods and services. This relationship is the law of supply. Higher prices make it worthwhile for additional providers to enter and for existing providers to expand output. Marginal cost-the cost of producing one more unit-often rises as capacity is stretched, so firms need higher prices to cover those rising extra costs.

Supply. Supply of a good or service is the quantity available for purchase. It depends on production capacity and resources and on the price that can be charged.

Law of supply. Ceteris paribus (all other things held constant), a higher price leads to a higher quantity supplied for most goods and services. The supply curve slopes upward.

Ceteris paribus. A modelling phrase meaning "all other relevant factors held constant", used so that the effect of one variable (often price) can be isolated.

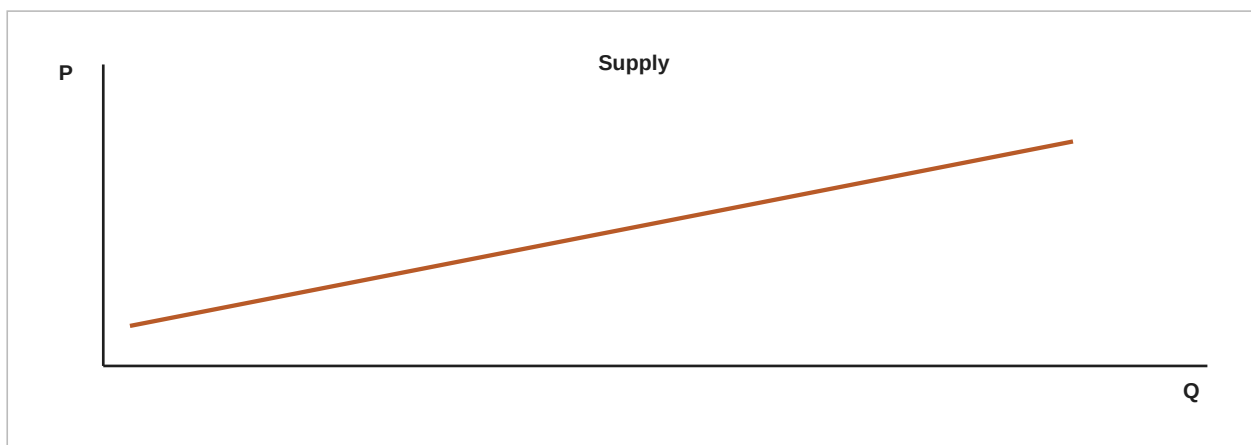


Figure 2. The supply curve - price on the vertical axis, quantity on the horizontal axis; the curve slopes upward as quantity supplied rises with price (*ceteris paribus*).

Graphically, price is usually placed on the vertical axis and quantity on the horizontal axis. The supply schedule traces pairs of price and quantity supplied. A change in price alone produces a movement along the supply curve. A non-price factor that changes supply at every price—more suppliers, better technology, lower resource costs—shifts the whole curve left or right.

Supply relationship (*ceteris paribus*): As P up $\rightarrow$ Q_s up (law of supply). Movement along the supply curve: a price change alone changes quantity supplied. Shift of the supply curve: a non-price factor changes supply at every price.

Demand is the quantity of a good or service that customers are willing and able to buy at a given price. Usually the higher the price, the lower demand will be. Willingness to pay reflects the utility-or satisfaction-buyers expect. As price rises, more buyers drop out or buy less; as price falls, more buyers enter or buy more. Plotting those pairs produces a downward-sloping demand curve. *Ceteris paribus*, a higher price leads to a lower quantity demanded for almost all goods and services. That inverse relationship is the law of demand.

Demand. Demand is the quantity of a good or service that customers are willing and able to buy at a given price.

Law of demand. *Ceteris paribus*, a higher price leads to a lower quantity demanded for almost all goods and services. The demand curve slopes downward.

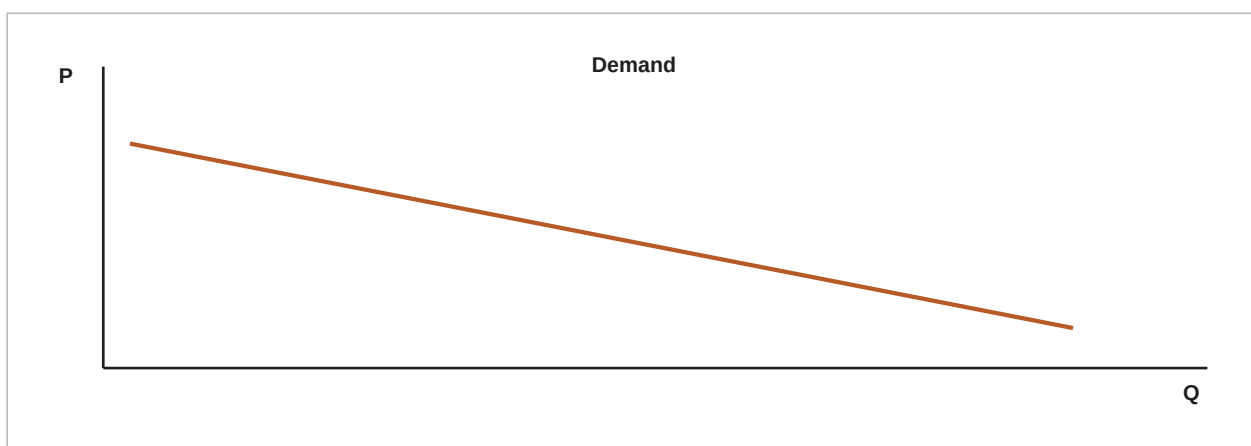


Figure 3. The demand curve - quantity demanded falls as price rises (*ceteris paribus*), so the curve slopes downward.

Demand relationship (ceteris paribus): As P up $\rightarrow Q_d$ down (law of demand). Movement along the demand curve: price change alone changes quantity demanded. Shift of the demand curve: income, tastes, complements, substitutes or other non-price factors change demand at every price.

If supply and demand are drawn together, they typically intersect at a certain price. At that point the quantity demanded equals the quantity supplied. That clearing price is the market or equilibrium price. There is neither surplus (supply exceeding demand) nor shortage (demand exceeding supply). At a price above equilibrium, quantity supplied tends to exceed quantity demanded and pressure builds for the price to fall. At a price below equilibrium, quantity demanded tends to exceed quantity supplied and pressure builds for the price to rise.

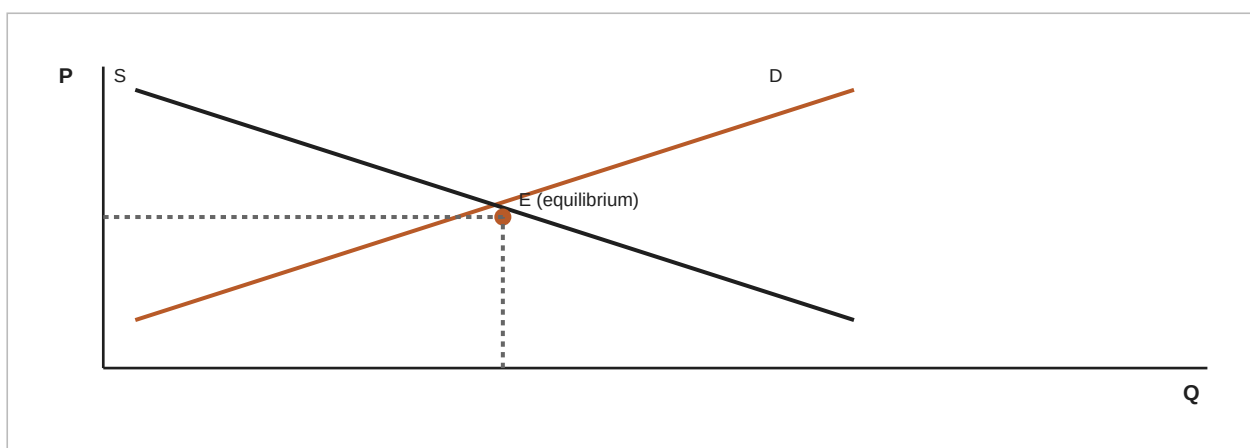


Figure 4. Market equilibrium - supply and demand intersect at the market (equilibrium) price, where quantity supplied equals quantity demanded.

Market equilibrium. Equilibrium (market) price is the price at which quantity supplied equals quantity demanded. There is neither surplus nor shortage at that clearing point where the curves intersect.

Equilibrium condition: At equilibrium: $Q_s = Q_d = Q^*$. If $P > P^*$, typically $Q_s > Q_d$ (surplus) and pressure for price to fall. If $P < P^*$, typically $Q_d > Q_s$ (shortage) and pressure for price to rise.

In the real world, factors other than the good's own price also affect demand and supply. Demand shifts when incomes change, when consumer preferences change, when complementary goods become more or less attractive, or when substitutes become more or less available. An increase in demand at every price appears as a rightward shift of the demand curve; a decrease appears as a leftward shift. Supply shifts when the number of suppliers changes, when technology improves, when resource prices change, or when providers' price expectations lead them to expand or leave the market. It is a common mistake to say "demand increased" when only price fell and quantity demanded rose along an unchanged curve. Quantity demanded responds to price; demand as a whole responds to non-price factors.

The same language illuminates one channel of inflation. If the quantity of money in an economy rises and demand for goods climbs while real output does not keep pace, general prices tend to rise—"too much money chasing too few goods". Raising interest rates makes borrowing more expensive, which can cool demand and ease inflationary pressure. That link between money markets and goods markets is a macroeconomic application of supply-demand thinking.

Worked example. Suppose tutoring equilibrium is €40 per hour with 1,000 hours traded each week. If incomes rise and more families can afford tutoring, the demand curve shifts right: at the old €40 there is now a shortage, so the new equilibrium price and quantity are both higher under a standard upward-sloping

supply. If software suddenly makes tutors more productive and lowers their costs, supply shifts right: equilibrium quantity rises and equilibrium price tends to fall.

2.7 Competition in the market

The intensity of competition in a market depends mainly on the number of suppliers and on how easily buyers can switch to substitutes. Market structure ranges from a single seller to many sellers of nearly identical products. Pure models help diagnose market power even when real markets mix features.

If there is only one supplier, the market situation is called a monopoly. Pure monopolies are rare in a market economy, but a firm can enjoy monopoly-like power in a local area if no close alternative exists nearby—for example a ski hut that is the only place offering food on a particular mountain, or a single concession stand in a theatre. Monopolists can set prices with more freedom, constrained mainly by the demand curve and by possible regulation.

Monopoly. A monopoly exists when there is only one supplier in a market. Pure monopolies are rare, but a firm can enjoy monopoly-like power in a local area if no close alternative exists nearby.

If there are few suppliers, each holding a relatively large share, the market form is an oligopoly. Telecom operators and large car manufacturers often illustrate oligopolistic structures. Rival reactions matter: one firm's price or product move typically triggers responses. Firms may compete fiercely on price and features, or they may be tempted to coordinate terms. Cartels—agreements that restrict competition—are generally illegal because competition law protects buyers. It is a common mistake to assume that cartel coordination is encouraged because it raises revenues for firms; policy supports competition for customers' benefit.

Oligopoly. An oligopoly is a market with few suppliers, each holding a relatively large share. Strategic interdependence means that one firm's moves typically prompt rivals to react.

Perfect competition is a theoretical benchmark with so many buyers and sellers that no single agent can influence price. Idealised assumptions include full information, free entry and exit, and no personal preferences—goods are replaceable (homogeneous). Perfect competition is rarely found as a complete package, yet some markets approximate it when products are nearly identical and rivalry is intense—certain agricultural markets, for example. Even with fewer sellers, competition can feel almost perfect if rivalry is very strong and products are standardised.

Perfect competition. Perfect competition is a theoretical benchmark with so many buyers and sellers that no single agent can influence price. Assumptions include full information, free entry and exit, and homogeneous goods.

Policy makers usually favour competition because it pressures firms to keep prices in check, innovate and serve customers. Barriers to entry, exclusive control of a key input, or network effects can reduce competition. Substitutes discipline prices even when a branded product looks unique: if switching is easy, monopoly power shrinks. Geography and travel costs matter: a firm that is not the only seller in a whole country can still hold local monopoly power.

Worked example. A remote valley has one grocery van that visits twice a week and no other store within an hour's travel—locally monopoly-like for many staples. A national mobile phone market with three large network operators is oligopolistic: a promotional tariff by one usually prompts matching offers. A large wholesale market for standardised wheat grades, with many growers and buyers, comes closer to perfect competition than either example.

Chapter 3

Focus on different types of businesses

Private households come in many forms-large families, singles, couples, with or without children-and so do businesses. The variety of businesses reflects differences in the factors of production they combine, the sectors in which they operate, their orientation towards profit, their size, and their geographic reach. Across all of those differences, every firm operates inside an environment of stakeholders whose interests can align or conflict.

This chapter trains the reader to classify businesses along those dimensions. The same enterprise can be described as tertiary, profit-oriented, micro and local, and stakeholder mapping completes the picture. Precise terminology supports clear thinking: factors of production, primary-secondary-tertiary sectors, profit orientation, SME size bands, local-national-international scope, and stakeholders.

3.1 Businesses combine different factors of production

A business is an entity that offers goods and/or services to customers. In order to create that offer it must combine factors of production-the resources used to generate output. Different industries emphasise different factors, which is one reason business landscapes look so varied. A vineyard leans on land; a consulting duo leans on labour, knowledge and entrepreneurship; a capital-intensive plant leans on machinery and finance. Most real firms use several factors at once.

The classical factors are labour, land and capital, joined by entrepreneurship as the organising and risk-taking function that brings the others together. Knowledge and technology sit alongside that set. Recipes, software, know-how and production methods can be decisive even when land holdings are modest. A logistics firm with clever routing algorithms may outperform a rival with more trucks but weaker systems. Technology does not replace the need for labour and capital; it reshapes how those factors combine and which skills earn a premium.

Factors of production. Factors of production are the resources used to create goods and services. Core categories include labour (human resources), land (natural resources), capital (machinery, plant, vehicles and financial resources), entrepreneurship (the organising and risk-taking function that combines other factors), and knowledge and technology.

Labour. Labour covers all human effort and skills applied in production-from manual work to highly specialised professional services.

Land. Land means natural resources: soil, minerals, water rights, locations and other gifts of nature used in production.

Capital. Capital includes produced means of production such as machines, buildings, vehicles and tools, as well as financial resources used to fund them.

Entrepreneurship. Entrepreneurship is the factor that brings land, labour and capital together, takes initiative and bears business risk in pursuit of an opportunity.

Factor mixes explain business diversity. Repair and tutoring start-ups often run on founder labour, knowledge, light capital and entrepreneurship. Food producers may combine extensive land, seasonal labour, equipment and agronomic know-how. Electronics manufacturers typically assemble large amounts of capital equipment, specialised labour, purchased materials and process knowledge. When classifying a firm, it helps to ask which factors dominate the value-creation story.

Factors are scarce, so combining them well is an economic decision in the sense of Chapter 2. Hiring one skilled engineer means not spending that wage bill elsewhere. Buying land ties up finance that could have funded inventory. Entrepreneurship is itself scarce: not everyone is willing or able to organise production

under uncertainty. It is a common mistake to treat capital as nothing more than cash in a personal bank account; in economics it includes physical means of production and financial resources tied to production. Entrepreneurship, likewise, is a factor of production-the organising and risk-bearing resource-not merely a personality trait.

Worked example. A hillside orchard uses land (orchard plots), labour (seasonal pickers and permanent caretakers), capital (tractors, storage and sorting equipment, working capital), entrepreneurship (owners who decide crops and carry sales risk), plus knowledge (pruning methods, pest management) and technology (irrigation controls). A city language school in rented rooms uses little agricultural land, but still uses location as a resource, relies heavily on teacher labour and curriculum knowledge, needs capital for rooms and platforms, and depends on founders' entrepreneurship to attract students.

3.2 Businesses operate in the primary, secondary and/or tertiary sector

Depending on what a business does, it contributes to one or more of three sectors of the economy. The three-sector model classifies activity by primary function. The primary sector extracts raw materials from nature-mainly farming, fishing, mining and forestry. The secondary sector transforms materials into manufactured goods such as cars, machinery, clothing, electronics and processed foods. The tertiary sector comprises services: distribution, banking, insurance, coaching, software support, hospitality, healthcare and education services, among others.

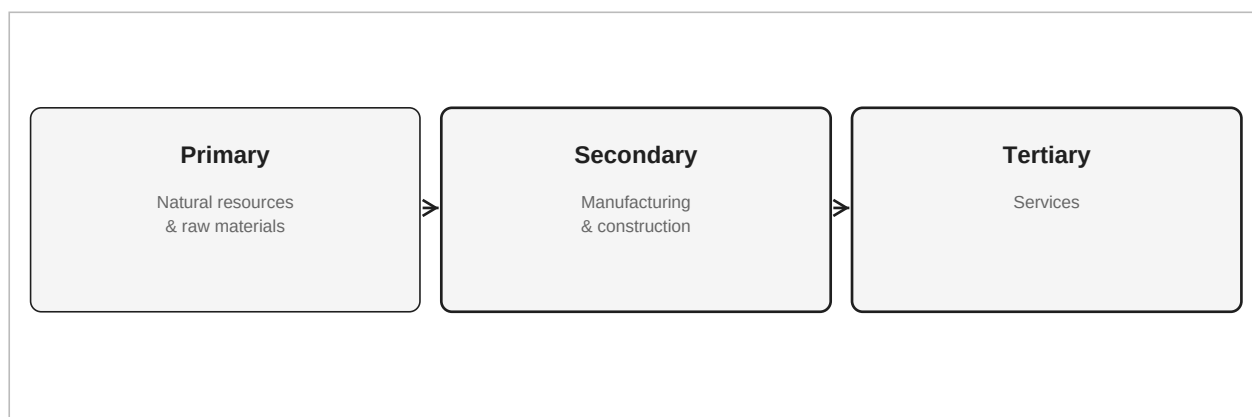


Figure 1. The three economic sectors - primary (extraction of raw materials), secondary (manufacturing) and tertiary (services), which together structure economic activity.

Primary sector. The primary sector covers extraction of raw materials from the earth, mainly farming, fishing, mining and forestry.

Secondary sector. The secondary sector transforms raw materials into goods through manufacturing.

Tertiary sector. The tertiary sector comprises services such as distribution, banking, insurance, coaching, software support, hospitality, healthcare and education services.

A single enterprise can straddle more than one sector-for example a dairy that farms, processes cheese and runs a farm shop-but classification usually follows the dominant activity. Emerging economies often depend relatively heavily on the primary sector. As countries become more economically developed, the tertiary sector usually becomes the largest share of output. In many highly developed EU economies, services can account for more than 70% of economic output, while the primary sector-though vital for food and materials-forms only a small percentage of measured output. A large service share of GDP does not mean food production is unnecessary; sector shares describe measured structure, not biological importance.

Gross domestic product (GDP). GDP is the total monetary value of final goods and services produced within a country's borders in a given period (usually a year). It measures overall economic

activity and, when adjusted for inflation, serves as an indicator of economic growth.

GDP is correlated with some well-being indicators such as health status and reported happiness, yet it is debated as a complete welfare measure. It may miss some income sources, say little about sustainability or distribution, and can rise after disasters because rebuilding spending counts as production even though people are worse off. Growth that lifts real GDP over time indicates an expanding economy; it does not automatically prove higher well-being in every respect.

Sector labels connect to the circular flow and to specialisation. Primary producers feed secondary processors; secondary goods move through tertiary distributors and financiers. International specialisation often follows sector strengths: some regions emphasise resource extraction, others advanced manufacturing, others financial or digital services. Do not assume that "services" means unimportant: logistics, banking and design enable industrial and agricultural value chains.

Worked example. A mining company extracting copper operates in the primary sector. A plant turning copper into cables is secondary. A utility that installs and maintains grid connections for households is tertiary. A retailer selling packaged food is tertiary even though food originated in farming; the retailer's main activity is distribution and customer service, not extraction. Manufacturing belongs to the secondary sector because it transforms materials; using raw materials does not make a factory primary.

3.3 Businesses can be profit-oriented or not-for-profit

Most businesses aim to continue over time. Sustaining operations usually requires that revenues cover costs and leave a surplus that can absorb shocks and fund reinvestment. Profit-oriented businesses explicitly pursue profit as a central goal: they intend revenues to exceed costs and expenses. That surplus rewards owners and investors for risk and can be ploughed back into capabilities that strengthen durability.

Not-for-profit organisations mainly aim to cover their costs while pursuing a social, cultural, humanitarian or environmental mission. They still need donations, fees, grants or membership income: without sufficient inflows they cannot pay staff or deliver programmes. Surpluses, when they occur, are beneficial because they can expand impact, but the organising purpose is mission coverage rather than maximising owners' financial returns. It is a common mistake to read "not-for-profit" as "never records a surplus" or "needs no revenue". Surpluses support the mission; donations and other inflows remain necessary to cover costs.

Aspect	Profit-oriented business	Not-for-profit organisation (NPO)
Primary aim	Earn profits (revenues above costs) as a central goal	Cover costs while pursuing a mission
Use of surplus	Reward owners/investors for risk; reinvest to strengthen the firm	Reinvest to enhance services or projects; not distributed as private profit goal
Need for revenue	Sales and other commercial income required	Donations, fees, grants or membership income still required
Success focus	Financial return alongside viability and competitiveness	Mission delivery and financial sustainability
Typical examples	Private firms, commercial partnerships, profit-seeking corporations	Charities, humanitarian NGOs, many cultural and environmental associations

Table 1. Profit-oriented businesses and not-for-profit organisations compared

Profit-oriented business. A profit-oriented business aims to earn profits-revenues higher than costs and expenses-so that owners are rewarded for risk and the firm can reinvest to remain viable and competitive.

Not-for-profit / non-profit organisation (NPO). A not-for-profit organisation mainly aims to cover its costs while pursuing a mission. Any surplus is reinvested to enhance services or projects rather than distributed as a private profit goal.

Hybrid situations appear in practice: social enterprises may earn commercial revenue while prioritising social outcomes. For classification, read the stated purpose. If the description stresses investor returns and market surplus as the driving aim, treat it as profit-oriented. If it stresses cost coverage and mission, with surplus recirculated into the cause, treat it as not-for-profit. Legal details of charity law vary by country; the conceptual contrast is what matters here.

Profits matter to stakeholders beyond owners: lenders watch repayment capacity; employees may link stability to financial health; communities may benefit from durable employers. Conversely, an NPO that chronically underfunds operations can fail its beneficiaries. Sound management of revenues and costs is therefore common ground-even when ultimate goals differ.

Worked example. *A private language school that expands only when expected profits justify investment is profit-oriented. An international relief organisation that raises donations to provide emergency aid, covering overheads and reinvesting any excess into more programmes, is not-for-profit. Both must manage cash carefully; only the first treats owner profit as a core success metric.*

3.4 Businesses come "in all sizes": large and small

Businesses range from micro outfits to large corporations. In the European Union, about 99% of all businesses are SMEs (small and medium-sized enterprises), sometimes discussed together with micro firms as MSMEs. Size categories matter practically: they affect access to finance programmes, eligibility for support schemes, and often the complexity of legal and accounting obligations.

SME (small and medium-sized enterprise). SME is the umbrella for smaller firms below large-enterprise thresholds. EU practice further distinguishes micro, small and medium-sized companies using staff headcount together with turnover or balance-sheet total criteria.

The European Commission's widely cited size bands combine staff headcount with either turnover or balance-sheet total. Micro enterprises have fewer than 10 staff and turnover or balance-sheet total of at most €2 million. Small enterprises have fewer than 50 staff and at most €10 million on either financial measure. Medium-sized enterprises have fewer than 250 staff and turnover of at most €50 million or a balance-sheet total of at most €43 million. Firms above those ceilings are treated as large. Formal classification can involve additional ownership and linkage rules; for introductory purposes the headline thresholds and their practical importance are enough.

Company category	Staff headcount	Turnover	OR Balance sheet total
Micro	< 10	≤ €2 m	≤ €2 m
Small	< 50	≤ €10 m	≤ €10 m
Medium-sized	< 250	≤ €50 m	≤ €43 m
Large	Above SME ceilings	Above SME ceilings	Above SME ceilings

Table 2. EU size categories of businesses (SME definition)

A two-person repair service that hires two assistants remains a micro firm by headcount. A manufacturer with around 10,000 employees is large. Between those poles sit tens of thousands of small and medium firms that form the backbone of employment in many regions. Size is not destiny: micro firms can be highly specialised and profitable; large firms can be inefficient. Size does, however, change management stretch, access to capital markets, and regulatory load. It is a common mistake to treat "SME" as a synonym of "micro only": medium-sized firms are still SMEs. Likewise, the claim that almost all EU businesses are SMEs refers to the count of enterprises, not automatically to every economic aggregate; large firms can still dominate revenues in some markets.

Accounting rules often ease burdens for smaller entities: simpler methods may be allowed when calculating profits or losses. That relief recognises limited administrative capacity. Conversely, larger entities face richer disclosure expectations because more stakeholders depend on their information.

Chapter 6 develops accounting; here it is enough to note that size and reporting obligations travel together.

Worked example. Studio A has 6 employees and €1.2 million turnover: micro. Studio B has 40 employees and €8 million turnover: small. Factory C has 180 employees, €40 million turnover and a €30 million balance-sheet total: medium-sized. Group D has 8,000 employees worldwide: large. Eligibility for an SME innovation grant would typically open to A-C, subject to full legal tests, and exclude D. A firm with 200 employees can still be medium-sized under the headcount idea; it is not automatically large.

3.5 Businesses may be local, national or international

Geographic reach is another dimension of business variety. Some firms serve only a neighbourhood; others cover a whole country; others produce or sell across borders. Expansion widens opportunity but stretches logistics, regulation and cultural competence. Classification follows where customers and operations are mainly located.

A local or regional business operates in a limited area. Most customers live nearby; the firm does not serve the national or international market in a material way. Challenges often include undercapitalisation-own funds are limited and external finance is hard to obtain-and acquiring a substantial customer base in a small catchment. Choosing a good location is critical because customers may not travel far.

Local (regional) business. A local or regional business operates in a limited area. Most customers live nearby; the firm does not serve the national or international market in a material way.

A national business serves the home-country market but not international markets-for example an insurer or bakery chain present across one country yet not abroad. National firms still choose locations carefully, but they must also design distribution so goods and services reach other domestic regions. Supply chains lengthen compared with purely local trade.

National business. A national business serves the home-country market but not international markets.

International or multinational businesses make and/or sell goods or services in more than one country. Internationalisation often responds when home markets feel too small or when cost and know-how advantages appear abroad. Challenges multiply: longer supply chains, different legal and economic systems, languages and cultures, and possibly multiple currencies. As more firms operate internationally, economic life becomes more globalised-globalisation referring to deepening cross-border integration of markets, production and information.

International (multinational) business. An international or multinational business makes and/or sells goods or services in more than one country, facing longer supply chains and cross-border legal, cultural and sometimes currency differences.

Scope can change over time. A café may begin local, franchise nationally, then license a brand internationally. Online channels can blur labels: a tiny team may ship worldwide from one warehouse. For classification, emphasise where value is mainly made and sold according to the facts, not marketing rhetoric about being "global" without evidence. Importing a few supplies from abroad does not automatically make a corner shop an international business if customers and operations remain purely local. Serving an entire single country without foreign markets remains national, not international.

Worked example. A neighbourhood bike workshop whose clients all live within a few kilometres is local. A supermarket chain with stores in every major city of one country but none abroad is national. A components manufacturer with plants in Europe and Asia that sells to industrial clients worldwide is international. Each faces a distinct logistics and regulatory map.

3.6 Businesses operate in an environment - stakeholders are important

Whatever their sector, size or reach, businesses sit inside an environment of other people and organisations. Anyone who is affected by a firm's activities or who has an interest in what the firm does is a stakeholder. Globalisation expands that circle: distant suppliers, overseas customers and international NGOs may all become relevant. Rising environmental awareness also raises expectations that firms protect natural resources rather than treat nature as a free dump.

Stakeholder. A stakeholder is anyone who is (potentially) affected by a business's activities and/or has an interest in what the business does.

Stakeholder group	Typical interest / concern
Owners / investors	Return on capital for risk taken; value growth; sometimes mission and social contribution
Managers	Income, career identity, ability to implement strategy; may or may not also be owners
Employees	Wages, job security, meaningful work; shared values with the organisation
Customers	Useful, reliable offers at acceptable prices; ongoing relationship
Suppliers	Timely orders, fair payment, stable future demand
Government	Tax compliance, observance of rules, contribution to public aims
Communities (local, national, international)	Jobs, congestion, pollution, cultural sponsorship, licence to operate
Natural environment	Sustainable use of resources; reduced emissions, waste and ecological harm

Table 3. Main stakeholder groups of a business and typical interests

Owners invest capital and seek returns for risk; they may also care about problem-solving and social contribution. Managers and employees depend on the firm for income and often for identity; the firm depends on their competence and commitment. Shared values and objectives support short- and long-run success. It is a common mistake to treat stakeholders as identical with shareholders: employees, suppliers, customers, communities and government also qualify when they are affected or interested.

Suppliers and customers form mutual dependencies. Late or poor-quality deliveries halt production; unpaid invoices damage suppliers. Customers expect useful offers; without customers, revenue collapses. Governments collect taxes and set rules; communities experience jobs, congestion, pollution or cultural sponsorship. Ignoring a powerful stakeholder can threaten licence to operate even when short-term profits look fine.

Environmental responsibility requires more than slogans. Claiming care without evidence-"greenwashing"-erodes trust. Credible firms report concrete impacts and improvements (energy, waste, water, emissions) and change processes accordingly. Stakeholder interests can conflict: owners may want higher margins while employees want higher wages and neighbours want quieter nights. Management includes negotiating those tensions rather than pretending they vanish.

Stakeholders are only one success factor among several. Legal structure, financial structure, market awareness, cost control and profitability also matter-and later chapters expand those themes. Classification of businesses by factors, sector, profit orientation, size and geography remains incomplete without seeing the web of interests around every firm.

Worked example. A mid-sized food processor lists stakeholders: family owners (dividends and continuity), plant managers, production staff, farmers who supply milk, supermarket buyers, the municipal government (permits and taxes), nearby residents (noise and traffic), and the river ecosystem affected by wastewater. A plan to run night shifts may please owners and some customers needing faster delivery, yet upset residents and tired staff-showing conflict that must be managed explicitly.

Chapter 4

Forms of business ownership and sources of finance

As soon as businesses exchange goods and services with customers, a legal question arises: who is the other party to the contract - the business as a separate entity, or the people who own and run it? The answer matters when contracts are closed, when capital is invested, and when disputes occur. It depends on the legal structure chosen for the firm. That choice shapes ownership, management, liability for debts and obligations, the taxation of profits, continuity if an owner leaves or falls ill, and the range of financing options available.

Founders must decide whether one person or several will own the business, whether owners will also manage day-to-day operations, whether liability is limited or unlimited, and how capital will be raised for start-up, operations and growth. Some owners wish to develop objectives, organise activities and monitor performance themselves. Others prefer to invest capital and leave management to appointed directors. Finance is needed at every stage: initial assets and recurring expenses must be paid for; some structures impose minimum capital rules; later expansion often requires funds beyond cash from sales alone. Legal form both constrains and enables those options, which is why growing firms sometimes change structure as their financing needs evolve.

Although names and formal rules differ across countries, the same basic ownership patterns appear widely: sole proprietorships (sole traders), partnerships (general and limited), and corporations (including public and private limited companies). German-speaking jurisdictions use labels such as Offene Gesellschaft (OG), Kommanditgesellschaft (KG), Aktiengesellschaft (AG) and Gesellschaft mit beschränkter Haftung (GmbH). This chapter develops those forms and then classifies the main sources of finance - internal and external, equity and debt, short-term and long-term - before examining the criteria used when choosing among them.

4.1 Sole proprietorship / sole traders

A sole proprietorship - also called a sole trader - is a business owned by one person who also manages and runs the business. It is easy to establish, especially for small firms, because there are typically no minimum capital requirements to get started. The business is not a legal entity of its own. In the eyes of the law, the firm and the owner are the same party. Contracts are concluded in the owner's name; debts and obligations attach to the owner personally. Because there is no separate legal personality, profits are reported on the owner's personal income tax statement. The owner pays tax on profits earned from the business as personal income, not as a corporation. That simplicity attracts many micro-businesses, yet it also means that personal and business financial affairs remain closely linked for both tax and liability purposes.

Sole proprietorship (sole trader). A business owned by one person who also manages and runs it. The firm is not a separate legal entity, so profits are taxed as the owner's personal income and liability is unlimited.

Unlimited liability. The owner is liable for all debts and obligations of the business. Private assets are also at stake if business assets cannot cover what creditors are owed, or if private assets were pledged as collateral.

Management largely depends on the sole proprietor. On the one hand, the owner can make all management decisions without having to consult others, which brings speed and clarity of control. On the other hand, continuity problems may occur if the sole proprietor wants to retire or suffers a long-term illness: knowledge, relationships and decision rights are concentrated in one person. If support is needed,

the proprietor can hire personnel, yet it remains the owner's task to take the most important management decisions and to bear the risk. Hiring employees does not transfer unlimited liability away from the proprietor; staff can support operations, but the owner still remains personally liable for business debts.

Available financial funds for a sole proprietorship mainly depend on the financial capabilities of the owner. If the proprietor lacks funds, it will be difficult to set up the business. Most sole proprietors invest their own savings. If extra money is needed, they may seek investors and/or banks. From the business's point of view, the owner's investment and funds from outside investors and creditors are external sources of finance. Banks and other financial institutions offer short-term credit (duration less than a year) and long-term credit. The creditor provides money for an agreed period, normally repayable with interest. Creditors often require assets as collateral, especially for long-term facilities; long-term bank loans are frequently secured on land and property (a mortgage). The most common short-term forms for small firms are a bank overdraft and trade credit. An overdraft is flexible: once the account is in place, the proprietor can draw funds when needed and pays interest only when the account is overdrawn. Trade credit arises when suppliers allow a payment period rather than requiring immediate settlement. All kinds of credit - short-term as well as long-term - are liabilities for the sole proprietor.

Once the business generates revenues, internal sources of finance may become available. If revenues exceed expenses, profit can be retained and reinvested (unless withdrawn by the owner). Sale of assets that are no longer needed is another internal option. Internal funds matter because no interest or similar financing charges need be paid for them. Still, retained profit is only available after the business has begun to earn it, and withdrawing all profit as personal drawings leaves nothing to reinvest. Ease of establishment and the absence of a minimum capital rule must not be confused with limited liability: sole proprietorships are simple to start precisely because they are unincorporated, and that same feature means that private assets remain exposed.

Worked example. *A neighbourhood bakery is set up by one baker who invests €12,000 of personal savings in ovens and fittings and opens a current account with a €3,000 overdraft facility. Flour and packaging are bought on 30-day trade credit. In year one the bakery earns €28,000 profit before drawings; the owner withdraws €20,000 for living costs and leaves €8,000 in the business. The €8,000 is an internal source of finance. If sales collapse and unpaid supplier invoices and the overdrawn account cannot be cleared from business cash, creditors can claim against the baker's private assets - unlimited liability in practice.*

4.2 Partnership

If two or more persons jointly found a business, the business is called a partnership. Partners need to set up a partnership agreement in order to settle rights and responsibilities as well as the division of profits and losses. A well-drafted agreement typically specifies each partner's percentage of ownership, how profit and loss are shared, the terms of the partnership, decision-making rules, dispute resolution, and what happens if a partner leaves, dies or wants to transfer an interest. The firm remains unincorporated: it is not a legal person separate from the partners. Personal taxation typically follows each partner's share of profit.

In a general partnership all partners have equal rights, liabilities and responsibilities (unless the agreement allocates roles differently while still treating them as general partners for liability). They can share tasks and specialise. In difficult situations they can exchange ideas and, potentially, make better decisions than a single owner working alone. In Austria this form is called Offene Gesellschaft (OG). Each partner in a general partnership is solely liable for all debts of the business (unlimited liability). That means a creditor can pursue any general partner for the full amount if business assets are insufficient - not merely for that partner's ownership percentage. Private assets of the partners may therefore be reached. Unlimited liability is the counterpart of the informal, flexible structure: partners gain shared control and specialised labour but accept personal exposure. Ownership percentages affect profit shares; they do not cap personal exposure for general partners.

General partnership. A partnership in which partners jointly own and manage the business and each partner has unlimited liability for the debts of the firm. In Austria this form is called Offene Gesellschaft (OG).

Limited partnership. A partnership with at least one partner who is not involved in management and whose liability is limited to the amount contributed to the business. In Austria this form is called Kommanditgesellschaft (KG).

In a limited partnership there is at least one partner who is not involved in the management of the business and whose liability is limited to the capital contributed. Active (general) partners typically retain unlimited liability and management rights; the limited partner provides finance and shares profits as agreed but does not run the firm. Crossing the management boundary can put limited-partner status - and limited liability - at risk under typical legal frameworks. Available financial funds for partnerships resemble those of sole proprietors, with an important scale difference: two or more partners can often invest more savings and raise more funds than a single owner, and may offer more private assets as collateral when seeking loans. Share issues on a stock exchange are not available to ordinary partnerships; equity comes from partners' capital contributions and any new partners admitted under the agreement.

***Worked example.** Two consultants found a general partnership (OG). Partner A contributes €20,000 and Partner B €10,000; they agree to share profits 60:40 after salaries for work done. The firm borrows €40,000 from a bank for office renovation. When the firm cannot repay and business assets are exhausted, the bank may claim the full remaining debt from either partner - not only 60% from A and 40% from B. If instead they had formed a KG with an investor as limited partner contributing €25,000 and taking no management role, that investor's loss would normally be capped at €25,000, while the managing partners would still face unlimited liability.*

4.3 Corporations

Corporations are businesses that are legal entities of their own. As legal persons they have the same rights and obligations, in business life, as natural persons: they can own land and property, hire people, close contracts, sue and be sued in their own name. The people who found the corporation and own a share of the business need not be involved in management, and the managers need not own a share. Corporations are more difficult to set up than sole proprietorships or partnerships, but shareholders' liability is usually limited to the amount invested when buying the shares. Limited liability means that shareholders risk only the money paid (or unpaid but committed) for their shares; creditors cannot normally claim shareholders' private assets for company debts.

Corporation (legal person). An incorporated business with separate legal personality. Ownership is divided into shares; shareholders' liability is typically limited to the capital they invested.

Limited liability. Shareholders risk only the money paid (or unpaid but committed) for their shares; creditors cannot normally claim shareholders' private assets for company debts.

Shareholders are not necessarily managers. Shareholders provide share capital but are neither obliged nor ordinarily entitled to run operations themselves. The corporation is managed by a board of directors - persons elected by the shareholders to make major business decisions and to represent shareholders' interests. The highest-ranking manager is often called the Chief Executive Officer (CEO). Other typical board roles include Chief Financial Officer (CFO), Chief Operating Officer (COO), Chief Information Officer (CIO) and Chief Marketing Officer (CMO). Separation of ownership and control is a defining feature of the corporate form: investors may specialise in risk-bearing capital while professional managers specialise in decision-making. Providing share capital does not entitle every investor to run operations; conversely, managers need not be shareholders.

Corporations usually have more options to raise funds than sole proprietors and partnerships. Their financial funds mainly comprise share capital as well as loans and credit. Share capital is the capital of the corporation divided into shares (also called stock). If share capital equals €1,000,000 and is divided into 100,000 shares, each share represents €10 or 0.001 per cent of the share capital. Persons who buy

shares become shareholders. Shares can be bought when initially issued by the corporation or later from another shareholder. If all 100,000 shares are sold at that issue price, the corporation gains €1,000,000 as share capital. Large amounts can thus be raised. Share capital is usually not redeemed by the company; it is long-term or even permanent capital.

A corporation's stock can but need not be listed on a stock market. A stock exchange is a regulated financial market where shares and other securities (for example bonds) can be bought and sold. Listed corporations must meet listing requirements. Shares are introduced at an issue price - an initial public offering (IPO) when first listed publicly - and subsequent prices are then driven by demand and supply. Rising demand may reflect expectations of profits, growth in market share or successful new products, and is also influenced by economic growth, interest rates and inflation. Critically, an increase in share prices after issue has no additional financing effect for the issuing corporation; the beneficiaries of secondary-market price rises are the selling shareholders only. Only issuing (or selling newly authorised) shares brings cash to the firm; secondary trading redistributes value among shareholders.

People invest in shares for several reasons: supporting a business they believe in; seeking dividends (parts of profit paid to shareholders, with no automatic obligation to pay them every year); hoping for capital growth if prices rise; exercising voting rights at the stockholders' meeting (typically attached to common stock); and treating shares as a claim on real economic activity that may hold value better under inflation than some cash holdings. Preferred shares may offer higher dividends with weaker or no voting rights, depending on the terms. Metrics of interest to investors include share price paths, market capitalisation (outstanding shares $\times$ current market price), dividend yield (dividend relative to price), and the price-earnings (P/E) ratio (share price relative to earnings per share).

Naming conventions differ by jurisdiction. In the United Kingdom and some Commonwealth countries, the public limited company (PLC) is a familiar corporate form. In the USA one often sees Inc. (Incorporated) or Ltd (Limited). In German-speaking countries the Aktiengesellschaft (AG) is the classic share corporation; Austria requires a significant minimum capital to found an AG (commonly cited as €70,000). Another transnational form is the European Company (SE, société européenne), similar to an AG/PLC and governed by EU law in member states. Besides bank loans and other credit available to smaller firms, large corporations can issue bonds: a loan relationship between investors as creditors and the corporation as borrower, with agreed repayment and interest. Bond interest is often lower than bank rates for comparable large volumes, which can make long-term investment in plant and infrastructure more affordable - without diluting ownership the way a new share issue does.

Private limited companies are corporations and legal persons with limited liability, but their shares are usually not sold to the general public on a stock exchange; instead they are offered among existing owners or tightly controlled parties. Examples include the LLC in the USA, the private company limited by shares in the UK, and the Gesellschaft mit beschränkter Haftung (GmbH) in Austria and Germany. Austrian syllabus figures commonly cite a minimum capital of €35,000 for a GmbH - lower than for an AG, still a barrier compared with a partnership, and without the public flotation path of a listed AG.

Worked example. A manufacturing corporation issues 200,000 shares at €25 each and raises €5,000,000 of share capital at IPO. Two years later the shares trade at €40 on the exchange. The market capitalisation is $200,000 \times €40 = €8,000,000$, but the corporation does not receive the €15 per-share price rise: that accrues to investors trading among themselves. Separately, the same firm issues a €2,000,000 bond at 4% interest to build a plant. Bondholders remain creditors, not owners; failure to pay interest and principal can force restructuring, whereas equity holders' ordinary downside remains limited to their share investment.

4.4 Summary: Overview of forms of business ownership

The forms of business ownership can be summarised by contrasting unincorporated and incorporated businesses. Unincorporated businesses are not legal entities of their own; incorporated businesses are legal persons. That single distinction drives differences in liability, taxation mechanics, continuity, and how ownership and management can be separated. Multiple owners alone do not create a legal person; incorporation does. Personal income tax treatment for a sole trader does not imply that the business is a

corporation - separate legal personality is what makes corporate tax and limited liability frameworks available.

Sole trader	One owner	Unlimited liability
Partnership	Two+ owners	Shared / unlimited*
Corporation	Shareholders	Limited liability

*Limited partners have limited liability in a limited partnership.

Forms of business ownership: unincorporated versus incorporated structures.

Among unincorporated businesses, owner(s) and manager(s) are typically the same person(s). With one owner, the structure is a sole trader / sole proprietorship. With more than one owner, the structure is a partnership - either a general partnership (all partners typically manage and face unlimited liability) or a limited partnership (with at least one limited partner outside management). Among incorporated businesses, owners and managers need not be the same persons: shareholders provide share capital while directors and executives run the company. This group includes public-style corporations such as AGs/PLCs and private limited companies such as GmbHs.

Criterion	Sole trader	Partnership	Corporation
Legal personality	None - firm identical with owner	None - unincorporated	Separate legal person
Owners	One person	Two or more partners	Shareholders
Management	Owner manages	Partners manage (limited partners do not)	Board / directors; owners need not manage
Liability	Unlimited	Unlimited for general partners; limited for limited partners (to contribution)	Limited to capital invested in shares
Taxation of profits	Personal income tax of owner	Typically personal tax on partners' shares	Corporate framework (company and/or shareholders)
Continuity	Fragile if owner retires or falls ill	Depends on agreement and partner succession	Stronger - legal person continues
Typical finance	Owner savings, overdraft, trade credit, bank loans	Partners' capital, loans, overdraft, trade credit	Share capital, loans, bonds (larger firms)
Setup difficulty / minimum capital	Easiest; usually none	Agreement required; usually none	Harder; AG ≈ €70,000, GmbH ≈ €35,000 (Austria)

Comparison of sole trader, partnership and corporation

Reading the overview as a decision path helps: start from how many owners and whether they want limited liability; then ask whether share transferability and public capital markets are needed; finally check minimum capital, disclosure and governance costs. Firms often begin unincorporated and incorporate later when risk, capital need or investor entry make separate legal personality and limited liability worth the

formalities. In complexity and financing reach, the usual ladder runs from sole trader to partnership to private limited company to public corporation.

Worked example. Three founders compare options for a logistics start-up needing €80,000 of vehicles. As a sole trader, one founder would own everything, face unlimited liability, and fund mostly from personal savings and a bank loan. As an OG, all three manage and share unlimited liability while pooling capital. As a GmbH (meeting minimum capital rules), all three take shares, liability is limited to invested capital, and they appoint managing directors - who may or may not be shareholders. The GmbH cannot yet float on an exchange; that would require moving toward a public corporate form and listing requirements later.

4.5 Overview of sources of finance

Every business needs financing - cash and credit that fund assets and cover expenses when sales receipts are insufficient or mistimed. Start-up requires money for equipment, inventory, rent, energy and insurance; ongoing operations and growth require further funds even when sales exist, because cash inflows and outflows rarely match perfectly. The balance sheet reveals which sources a business has already used (see Chapter 6 on accounting); large businesses are obliged to prepare such statements. For analysis, sources are classified along several overlapping dimensions: equity versus debt, internal versus external, and short-term versus long-term. Legal structure interacts with the menu of sources. Sole traders and partnerships rely heavily on owners' savings, bank credit, overdrafts and trade credit. Corporations add share capital as a major equity route and, when large enough, bonds as a large-scale debt route.

Equity finance. Funds that belong to the owners of the business (for example share capital and retained earnings). Equity is not a repayable loan with fixed interest, though owners expect returns via profit share, drawings or dividends.

Debt finance. Borrowed funds that create liabilities: bank overdrafts, trade credit, bank loans, bonds and similar instruments, normally carrying interest and a repayment obligation.

Category	Internal	External
Equity	Retained earnings; sale of surplus assets (cash release)	Funds from investors; share capital (new issues)
Debt - short-term (usually < 1 year)	-	Bank overdraft; trade credit; short-term bank loans
Debt - long-term (usually > 1 year)	-	Bank loans; loans from owners; bonds (larger corporations)

Common sources of finance by equity/debt and horizon

Equity finance may be internal or external. External equity includes funds provided by investors and share capital raised from owners who buy newly issued shares. Internal equity centres on retained earnings - profits kept in the business rather than distributed. Debt finance is external: creditors stand outside ownership. A bank overdraft lets a firm draw beyond the credit balance on its current account up to an agreed limit; interest is charged on the overdrawn amount and the facility is flexible for uneven cash flows. Trade credit is supplier-granted delay in paying for purchases - free if within the agreed period, costly if discounts are lost or late-payment penalties apply. Bank loans provide a fixed sum for an agreed term with scheduled repayment. Bonds package large-scale borrowing from multiple investors with contractual interest. Retained earnings require prior profitable operations and a decision not to distribute all profit; they are equity, not debt. Sale of assets that are no longer needed releases cash internally without creating new liabilities, but may reduce operating capacity if the asset is still useful. Trade credit and overdrafts are liabilities even when they feel routine: they must be repaid according to agreement and do not permanently fund long-lived assets without refinancing risk.

Worked example. A retailer finances a shop refit as follows: €50,000 retained earnings (internal equity), €30,000 new capital from a silent investor (external equity), a €20,000 three-year bank loan (long-term debt), a €5,000 overdraft for seasonal stock (short-term debt), and €8,000 of inventory purchased on

45-day trade credit (short-term debt). Selling an unused delivery van for €4,000 adds internal financing without interest. The firm's balance sheet will show the equity contributions and retained earnings on the owners' side and the loan, overdraft and trade payables among liabilities.

4.6 The choice of the source of finance

When several sources of finance are available, businesses choose among them using systematic criteria - not only the headline interest rate. The decision should reflect costs, the intended use of the funds (especially the duration of the need), flexibility, the effect on control and ownership, risk including gearing and insolvency exposure, and how the financing matches the life of the assets or projects being funded. Current financial structure also constrains what lenders and investors will still offer.

Costs comprise interest payments for loans and credit and administration costs for issuing shares or bonds (legal fees, underwriting, listing expenses). Equity may look cheaper in months without dividends, yet it can be costly in ownership dilution and required returns over time. Cheap short-term debt can become expensive if rolled over repeatedly at rising rates. Flexibility matters when cash needs fluctuate: an overdraft can expand and contract with activity, whereas a fixed term loan leaves idle cash costing interest if drawn too early, or a funding gap if the term ends before the project does. Control is affected when new shareholders enter - voting power and strategic influence may shift - while pure debt leaves ownership percentages unchanged but adds covenants and repayment pressure. Risk rises with the share of loan capital: repayment and interest are contractual; missing them threatens insolvency.

Matching (duration). Financing long-lived capital expenditures with long-term finance, and financing short-lived revenue expenditures with short-term sources, so repayment schedules align with the period over which benefits arise.

High gearing. A high proportion of loan capital relative to equity. Highly geared businesses may struggle to obtain more credit except at higher interest or with more collateral, and face greater insolvency risk because loans must be repaid.

If funds are used for capital expenditures - buying assets that will be used over some or many years - long-term finance is required. Revenue expenditures, such as buying materials used in near-term production, can safely be financed by short-term sources. Matching reduces the risk of having to refinance long assets under stress. Capital expenditure should not be routinely funded by permanent reliance on trade credit or overdrafts: short-term creditors can withdraw support precisely when the long asset still needs financing. A business that already has high gearing might face reluctant lenders, higher rates, and heavier collateral demands; such a firm should prefer internal sources and/or new equity investors or partners rather than stacking more debt.

Worked example. A workshop wants a €120,000 CNC machine with a ten-year useful life and €15,000 of steel for next month's orders. Financing the machine with a revolving overdraft would mismatch duration: the bank could demand repayment long before the machine pays for itself. A ten-year loan or lease matches better. The steel can sensibly sit on trade credit or a short-term facility. If the firm's loans already equal 75% of capital employed (high gearing), adding another large loan may be denied or priced punitively; retaining profit or admitting an equity partner may be the feasible path despite dilution concerns.

Chapter 5

Marketing

Producing a good or providing a service only makes sense if enough people want it and are willing to exchange something of value for it. Profit-seeking firms and non-profit organisations alike depend on sustained demand. Marketing is the management process - and for many organisations, a guiding philosophy - that identifies customers' needs and wants, shapes offerings around those needs, communicates benefits, sets prices customers will accept, and makes products available through distribution networks.

There is no single, universally accepted definition of marketing. The American Marketing Association describes it as the activity, set of institutions, and processes for creating, communicating, delivering, and exchanging offerings that have value for customers, clients, partners, and society at large. Other definitions adopt a similar approach: an ongoing management process aimed at identifying needs and wants, developing products and features accordingly, communicating benefits, pricing them acceptably, and making them available through a distribution network. Some writers treat marketing as more than a process - as a philosophy present throughout the business. Either way, marketing is about creating value for the customer. It rests on analysing the firm's strengths and know-how together with market research on customers and competitors.

Marketing is more than advertising, even though advertising is its most visible face. Critics note that promotion can push people to buy more than they need or can afford, and that continuous innovation can accelerate obsolescence and consumption. Supporters emphasise that good marketing creates genuine customer value: functional performance, additional features, image and the satisfaction users obtain. This chapter defines products in marketing terms, sets out marketing objectives, contrasts product orientation with market orientation, discusses responsibility and sustainability, develops market research including market-share measures, explains segmentation and targeting, and builds the marketing mix of product, price, place and promotion - including product portfolio tools such as the product life cycle and the Boston Consulting Group matrix.

5.1 What a product is

In marketing terminology, a product is every good and/or service that can be exchanged in order to fulfil the wishes and needs of customers. The definition is deliberately broad: tangible goods (a tool, a meal, a smartphone) and intangible services (repair, consulting, transport, education) both count when they can be offered in an exchange that satisfies customer needs. Bundles that combine goods and services - for example equipment sold with installation and maintenance - are still products in this sense. Restricting "product" to physical goods alone would miss a large part of modern markets.

Product (marketing). Every good and/or service that can be exchanged to fulfil customers' wishes and needs, including tangible items, services and combinations of both.

Customers can be other businesses or consumers (private households). Producer products are goods and services sold from one business to another (business-to-business, B2B). Consumer products are sold to consumers or private households (business-to-consumer, B2C). The classification hinges on the buyer's role and use, not on whether the item left a factory. The same physical printer is a producer product when purchased for an office workplace and a consumer product when purchased for a private home. Many firms operate in both B2B and B2C markets with different packaging, pricing, service levels and sales processes for each.

Producer product (B2B). A good or service sold from one business to another for use in production, operations or resale.

Consumer product (B2C). A good or service sold to private households or individual consumers for personal or household use.

Value for the customer may come from core functionality, additional features, brand image and the enjoyment or status of use. A mobile phone creates value because users can reach others from almost anywhere (core function), and because they can message, receive email and browse online (additional features). Owning a sought-after model may also signal expertise or lifestyle (image and emotional benefit). Marketing therefore designs and communicates the whole benefit package, not only the physical object.

***Worked example.** A firm sells industrial cooling units to factories (producer products, B2B) and compact air-conditioners to households (consumer products, B2C). It also sells annual maintenance contracts - pure services, still products in marketing terms. A technician who buys a unit for the company workshop is in a B2B exchange; the same model purchased for a living room is B2C. Marketing materials, warranty terms and price lists differ for the two customer types even though the hardware overlaps.*

5.2 Objectives of marketing

To fulfil customers' wishes and needs, a business must analyse the market(s) it wants to operate in and be clear about its marketing objectives. Objectives give direction to research, product design, pricing, distribution and promotion. They are typically interconnected: progress on one supports others, while neglect of one can undermine the rest.

Customer satisfaction is foundational. If customers are not satisfied with the product, they will not buy it again. Satisfied customers often become loyal customers who may repurchase and recommend the firm. Loyalty reduces acquisition cost over time and stabilises revenue. Measuring satisfaction through surveys, complaint rates or repurchase ratios turns a soft goal into something managers can track alongside sales.

Creating a unique selling proposition (USP) - also linked to product differentiation - makes the offering special or better than rivals in the eyes of customers. Differentiation can rest on a tangible product characteristic (durability, features, service intensity) or on how the product is promoted and perceived (image, association, storytelling). Building a brand supports USP formation by making identity and quality claims recognisable across places and over time. Without some form of differentiation, competition collapses into pure price comparison, which squeezes margins.

Unique selling proposition (USP). A distinctive benefit or perceived difference that sets a product apart from similar offerings and supports differentiation and customer preference.

Gaining and maintaining market share indicates relative importance and competitiveness versus rivals. Share objectives force the firm to define the market carefully: too broad a definition makes share look tiny; too narrow a definition can flatter performance. Maintaining or increasing sales generates the revenues needed to cover production and operating costs and to support profit. Profitability reimburses owners for invested capital and allows retention of earnings for reinvestment in product development, capacity or marketing itself. Higher sales often support higher profit, but not without limits: growth that requires heavy discounting, costly promotions or overstretched operations can expand revenue while eroding margin. Marketing objectives must therefore be set as a coherent set - satisfaction and USP feeding loyalty and share, sales feeding profit - with cost and capacity constraints made explicit.

***Worked example.** A regional bicycle brand sets objectives for the year: raise average customer satisfaction scores from survey panels, build a USP around lasting frames and free annual service, lift absolute market share in the urban-commuter segment from 8% to 10%, grow unit sales by 12%, and keep operating margin above 9%. Promotion emphasises the service USP rather than pure price cuts, so sales growth does not destroy profitability. Loyalty programmes track repeat purchases as a satisfaction and loyalty indicator.*

5.3 Product orientation versus market orientation

Over recent decades many businesses have moved from a product-oriented approach toward a market-oriented approach. A product-oriented business focuses on the product and its specifications first and later thinks about how to sell it - often believing that a good product sells itself, with advertising playing a supporting role. A market-oriented business first analyses customers' needs and wants and then tailors products to fulfil those requirements. Both may pursue similar high-level objectives (sales, share, profit), but their starting point and feedback loops differ. The product-oriented firm relies mainly on the quality of features to succeed; the market-oriented firm treats customer insight as the starting point of design.

Product orientation. An approach that prioritises production and product features first, then seeks to sell what has been made.

Market orientation. An approach that starts from customers' needs and wants and designs offerings to match them.

Market orientation has the advantage of anticipating changing needs and responding earlier than product-focused rivals. Suitability still depends on the product and the intensity of competition. Highly specialised engineering products with long development cycles may retain strong product focus, yet even they cannot permanently ignore market expectations. Neglecting the market risks building excellent objects that few customers want at the price offered.

Customer relationship management (CRM) aims to create a long-term relationship with customers. Data are kept so the firm can mail or email newsletters, coupons and product information that encourage repeat purchases. Using personal data has become a sensitive issue; careful, lawful and transparent use of personal data is indispensable. At the same time, many customers willingly share information through personal accounts and loyalty cards to obtain discounts and tailored offers. Firms that collect purchase histories can personalise assortments - a practical expression of market orientation - provided privacy expectations and rules are respected.

***Worked example.** A kitchen-appliance maker once designed ever-more complex mixers and then funded heavy advertising (product orientation). After stagnating sales, it interviewed home cooks, found demand for simpler, quieter machines that are easy to clean, and redesigned the line accordingly (market orientation). Loyalty-card data later showed which accessories were bought together, enabling CRM emails with relevant add-ons while offering clear opt-out choices to respect data sensitivity.*

5.4 The need for more responsibility and sustainability

It is often claimed that businesses do not only respond to what customers need or want, but also create wishes and needs by continuously developing new products and advertising them - sometimes in unethical ways. Continuous innovation can accelerate the perceived obsolescence of still-working goods. Easy checkout, credit and persuasive messaging encourage unplanned buying. Many people spend more than they can afford, or more than they intended, acquiring items they do not really need and will barely use. Overconsumption strains household budgets, generates waste and depletes materials and energy along supply chains.

Both businesses and consumers therefore need greater awareness of sustainable production and consumption and of the risks of consuming too much. Responsibility in this chapter is not a call to abandon marketing; it is a call to shape offers and messages so that informing and solving real problems outweighs manipulating impulse. Ethical advertising standards, honest claims and restraint in targeting vulnerable groups are part of that agenda.

Sustainable consumption and production. Patterns of making and using goods and services that meet needs while reducing unnecessary waste, resource use and harmful overconsumption.

Practical alternatives already appear in many markets. Repair and reuse extend product life: devices and appliances are fixed rather than replaced at the first fault. Consumers can choose higher quality that lasts, share or swap goods among friends, and rent instead of buy for occasional needs. In clothing, more people prefer durable garments or rental for special events rather than cheap items discarded after a few

wears. Firms can support these patterns with spare parts, repair services, refurbished lines and access-based business models - still marketing, but oriented toward longevity and use intensity rather than disposable volume alone.

Worked example. *An electronics retailer replaces "upgrade every year" campaigns with a repair desk, certified refurbished devices and transparent adverts about battery health. A fashion label launches a rental line for formal wear and accepts trade-ins for recycling fibre. Households that previously bought three cheap outfits per event now rent one high-quality set and return it. Waste falls; the firms still market - but around longevity and access rather than disposable volume alone.*

5.5 Market research

Market research provides information about existing and prospective customers, the potential buyers of a firm's products, about competition, and about the industry in general. Research data come from two kinds of sources: primary and secondary. Sound decisions on segmentation, targeting and the marketing mix rest on this evidence.

Primary information is gained by conducting an empirical study or by commissioning a market research institute. The study can be tailored to the business's specific interests: who customers are, which products they buy, what they think of those products, and whether they also buy rivals' goods. Methods include questionnaires administered to hundreds or thousands of respondents, personal or telephone interviews, and online surveys, followed by analysis and interpretation. Tailoring is a strength; cost is a weakness - especially for small businesses that cannot afford large primary programmes.

Primary market research. Original data collection designed for the commissioning business's questions (surveys, interviews, observation, experiments).

Secondary information uses research already done by someone else, often for another purpose. Government agencies, trade and industry associations and other organisations publish data that may be free or inexpensive. Secondary sources are faster and cheaper but usually more general and less tailored to one firm's precise questions. Existing published statistics are secondary; primary means new collection designed for the firm's questions. Many projects combine secondary scans with selective primary follow-up.

Secondary market research. Use of existing data collected by others (statistics, published reports, industry studies), typically cheaper but less specific.

Customer analysis is often central. Businesses try to learn who current and potential customers are (B2C consumers versus B2B buyers; also that buyer, user and influencer roles may differ - for example children influencing purchases paid for by parents). They study what customers do with products, to improve design around real use. They examine where customers buy, to evaluate distribution channels. They check when customers buy, to plan for seasonal swings and possible price differentiation over the year. They explore why customers choose one product over another, to guide development and market-share strategies.

Market research also measures markets. Market size (market volume) is the total sales of a product by all businesses in the market, expressed as value (for example euros) or quantity (units). Tracking how volume evolves helps firms judge their position and prospects. Market share is the proportion of that market held by a business, product or brand. Absolute market share relates one firm's sales to total market volume. Relative market share places the firm against its largest competitor, adding competitive context that absolute share alone does not provide.

Absolute market share: $\text{Absolute market share} = \frac{\text{sales volume of one business (or one brand)}}{\text{market volume}}$

Relative market share: Relative market share = market share of a business (or a brand) / largest competitor's market share

Market potential exceeds market volume when further customers could still be won. Sales potential for one firm exceeds its current sales volume when the firm can still gain from competitors and share in market growth. Distinguishing volume (what is sold now) from potential (what could be sold) prevents over- or under-estimating opportunities.

Worked example. A firm's sales in a defined market are €150,000; total market volume is €1,000,000. Absolute market share = $150,000 / 1,000,000 = 0.15 = 15\%$. The largest competitor holds 30% absolute share. Relative market share = $15\% / 30\% = 0.5$. Investors see that the firm is a meaningful but not leading player. If market potential is estimated at €1,200,000, about €200,000 of demand remains unrealised industry-wide; the firm's sales potential includes both conversion of that gap and gains from rivals.

5.6 Market segmentation and targeting strategies

Market research may reveal that characteristics such as age, gender, education, income, behaviour and interests help describe potential customers. Those characteristics can separate relatively homogeneous subgroups - market segments. Segmentation is worthwhile when segments are measurable (size, purchasing power), profitable, accessible through communication and distribution channels, and durable (not changing too quickly). The firm then selects which segment(s) to serve - targeting - and positions offerings so that the chosen groups see a clear match with their needs. Positioning builds an image or identity in the minds of target customers.

Market segmentation. Dividing a market into relatively homogeneous subgroups of customers who share relevant characteristics.

Targeting. Evaluating segment attractiveness and selecting one or more segments to enter with a tailored marketing approach.

Positioning. Creating a distinctive image or identity for the product in the minds of the chosen target market(s).

Base	Examples of variables	Typical use
Geographic	City, region, country; urban versus rural	Local reach, logistics, climate or language differences
Demographic	Age, gender, education, income, employment status	Purchasing power and life-stage offers
Psychographic	Lifestyle, values, attitudes (e.g. environmental concern)	Message tone and brand meaning
Behavioural	Usage intensity, occasions, loyalty, benefits sought	Pricing, service intensity, loyalty programmes

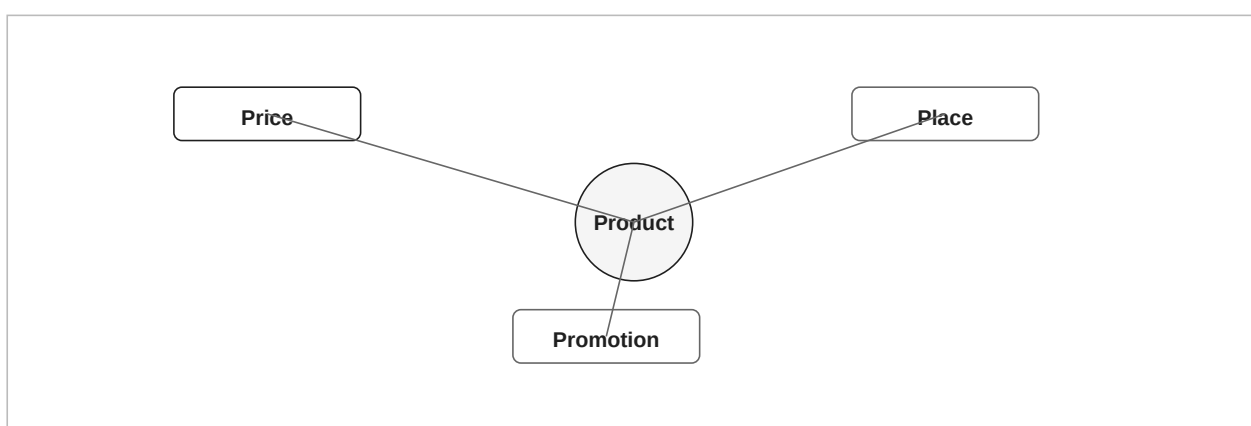
Common segmentation variables

Effective marketers often combine bases when a single cut is too crude. Mass marketing ignores segment differences and offers essentially the same product and promotion to everyone. It suits widely used items such as basic hygiene products or simple stationery and supports economies of scale: larger identical output can reduce cost per unit when some costs do not rise proportionally. The drawback is inflexibility when some customer groups change. Segment marketing (differentiated approaches) offers different products to one or more segments, focusing resources where strategic fit is strongest. Niche marketing goes further into particular subgroups within segments; many small specialised firms lead niches they understand deeply even if they could never serve a mass market. Mass marketing is not "more segmentation"; it deliberately ignores segments. Niche marketing involves more, not less, focus on particular subgroups.

Worked example. A refurbished-laptop seller defines a geographic focus on one metro region, demographics of employed adults and retirees, psychographics favouring lower-cost and lower-waste devices, and behaviour of occasional rather than heavy users. It targets small offices and home offices with limited IT budgets (segment) and within that specialises in buyers needing setup support (niche). It rejects mass marketing of identical devices nationwide because service capacity and brand promise cannot stretch that far. Positioning stresses "ready-to-use, supported, affordable and durable" rather than "latest model".

5.7 The marketing mix

A marketing mix is a harmonised blend of marketing tools that best meets the needs and wants of customers in the targeted market. It consists of four elements - product, price, place and promotion - the four Ps. Based on market research, the firm decides which goods and services to offer, at which price, where customers can obtain them, and with which message they are informed and persuaded. The Ps must fit together: a premium brand message fails if place is a discount channel and price undercuts quality cues, just as a convenience product fails if it is hard to find.



The marketing mix: product, price, place and promotion coordinated around market research.

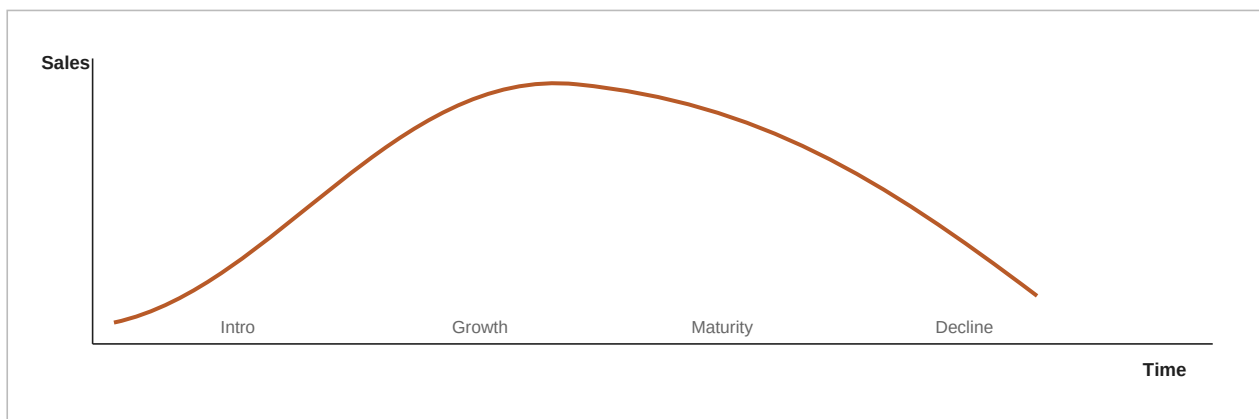
Marketing mix (four Ps). The coordinated set of Product, Price, Place and Promotion decisions used to serve a target market.

Product is at the heart of the mix and is usually the first major decision in serving a target market. Most firms offer a range rather than a single item. Closely similar items - for example several laptop configurations - form a product line. Several lines (laptops, monitors, printers, storage media) form a product mix. Brands - names, words, symbols or signs - distinguish products and firms, support USP and recognition, and signal consistent quality that customers can trust even when travelling or comparing unfamiliar outlets.

Product-mix strategies describe how the range evolves. A firm may specialise in one line or diversify by adding lines (increasing mix width). It may deepen a line with more variants, pursue line extension (new items inside an existing line, such as new flavours) or mix extension (entirely new lines, such as adding yogurts beside ice cream). Alteration or relaunch covers minor updates (packaging, colours). When relaunch is insufficient, major redesign or elimination (contraction) removes weak products or lines. These choices keep the portfolio aligned with changing needs and with capacity and brand limits.

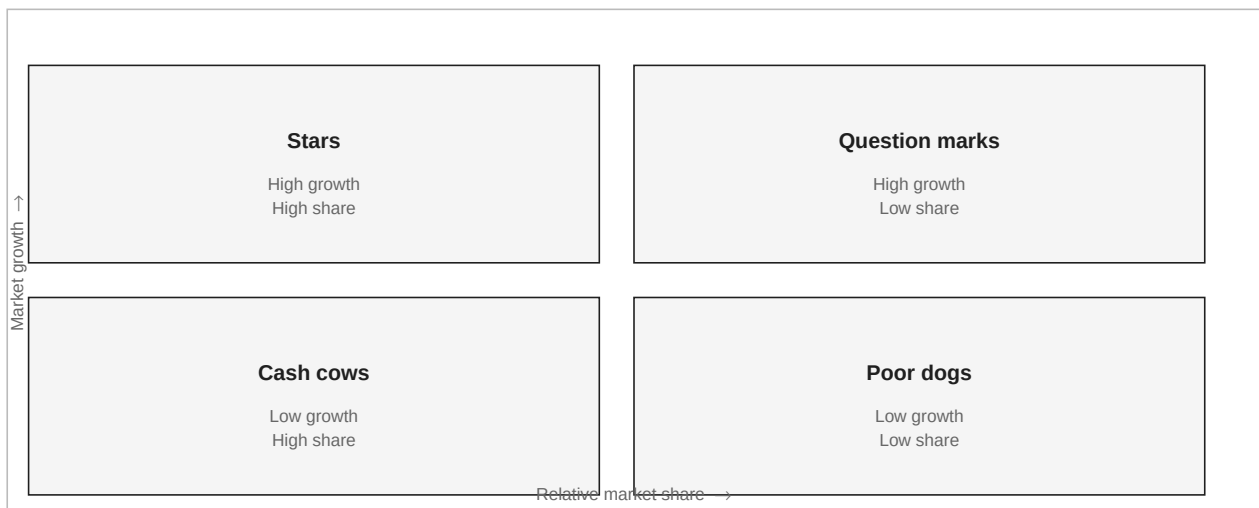
The product life cycle (PLC) is a theoretical model of stages over a product's life - introduction, growth, maturity and decline - that typically differ in sales volume and profit. Before launch there are development costs and no sales, so the business starts from a loss position. In introduction, sales begin but costs of promotion and possibly low introductory prices may keep profit negative or thin; toward the end of introduction a small profit may appear. During growth, sales rise faster than costs; average costs may fall with higher output and economies of scale; market share rises in a still-growing market. Profit usually strengthens and often peaks around maturity. Eventually competition, price pressure and saturation pull

the item into decline, where sales and profits fall. PLC length varies enormously - detergents may seem to mature indefinitely; fads may exit within a year.



The product life cycle: introduction, growth, maturity and decline.

The Boston Consulting Group (BCG) matrix links a product's relative market share to the growth rate of its market. A young item with low relative market share in a rapidly growing market is a "question mark". When share rises in a still-growing market the product may become a "star" that deserves continued investment in promotion and capacity. When market growth slows but relative share stays high, the product becomes a "cash cow": investment needs fall while revenues remain strong. Weak share in a slow market points toward "dog" status. A balanced portfolio blends stars, cash cows and selected question marks without too many dogs. Cash-cow revenues can fund stars (future cash cows) and question marks the firm believes in. BCG labels depend on relative market share and market growth together: high sales alone do not make a star if growth is low (likely a cash cow) or share is weak (question mark or dog).



The BCG matrix: stars, cash cows, question marks and dogs by relative share and market growth.

Typical PLC strategy shifts move from market development (introduction) to penetration (growth), defensive positioning (maturity) and efficiency or exit (decline). Extension strategies aim to delay decline by changing the mix or entering new markets. Ansoff's product-market matrix organises growth options: market penetration (existing product, existing market - lowest risk), product development (new product, existing market), market development (existing product, new market) and diversification (new product and new market - highest risk, but can spread long-run risk across lines).

Price is difficult to reset once customers form reference prices, so the initial strategy matters. Influences include costs (which must be covered in the long run, preferably with profit), competitors' prices for similar offers, and demand - how much customers are willing to pay. Cost-based pricing builds from cost; competition-based pricing watches rivals; demand-oriented pricing starts from willingness to pay. Total cost comprises fixed costs (independent of output in the relevant range, such as rent or insurance) and variable costs (rising with output). Contribution per unit equals selling price minus variable cost and feeds coverage of fixed costs. Break-even occurs where revenue equals total cost; break-even output equals fixed costs divided by contribution per unit (simplified one-product model).

Variable-cost-plus pricing adds a markup to variable cost (for example 50% on €480 yields a €720 selling price before tax). In competitive markets and contract bidding, distribution pricing similarly adds markup to variable cost. The absolute lower limit for a short-run accepting price is variable cost: any price above that still contributes to fixed costs, though chronically pricing below full cost destroys long-run viability. Price elasticity of demand compares the percentage change in quantity demanded with the percentage change in price. Demand is elastic when the quantity response exceeds the price change in percentage terms (elasticity magnitude greater than one); inelastic when the quantity response is smaller. Few substitutes and necessities tend toward inelastic demand; luxury and easily substituted goods toward elastic demand. Short-run and long-run elasticities can differ as habits and substitutes adjust.

Place (distribution) is easy to underestimate, yet a strong product fails if customers cannot find it where they expect it. Depending on the product, firms use distributors, wholesalers (who buy and resell to businesses), retailers (who buy and resell to consumers), agents or brokers (common in B2B), online selling, telemarketing, and formats such as vending machines or kiosks. Channel partners bridge factory and final buyer: logistics, local information and advertising support. A beverage producer, for example, cannot sell every unit directly to households nationwide and relies on wholesale and retail chains - or carefully designed direct-to-consumer online paths that still solve availability and delivery. In distribution discussions, "stock" often means inventory; elsewhere in finance chapters, "stock" may mean shares - context matters.

Promotion covers activities that inform potential customers and the wider public about the business, its products and their benefits. The promotion mix includes advertising on TV and radio, online and social media, newspapers and magazines, billboards, buses and trains; direct mailing; personal selling through salespeople; sales promotions; sponsorship of events, people or organisations; and public relations that build a favourable image and stakeholder relationships. Small budgets may emphasise websites, local ads, social presence, leaflets and word-of-mouth from satisfied customers. Whatever the tools, promotion must match the target segment and stay consistent with product quality cues, price positioning and place.

Mixing the four Ps means checking coherence for the chosen target. A niche refurbished-equipment offer typically combines a ready-to-use supported product, a price clearly below new equivalents yet above variable cost, low-inventory delivery-based place, and modest targeted promotion - not mass national TV. Premium new-product launches need matching price, selective place and heavier awareness advertising. Changing one P without adjusting the others can break the blend.

Worked example - contribution and break-even. Variable cost per unit €480, selling price €720, fixed costs €130,000. $\text{Contribution} = 720 - 480 = €240$. $\text{Break-even units} \approx 130,000 / 240 = 541.67$, so from the 542nd unit the firm moves into profit (simplified model). If price falls to €600, contribution is €120 and break-even doubles - illustrating why price cuts without volume gains can threaten profitability.

Chapter 6

Accounting - keeping record of business transactions

Every business generates a stream of transactions: purchases, sales, borrowings, repayments, wages, rents and many others. Bookkeepers record those transactions from supporting documents such as invoices and receipts so that the firm can later present a clear picture of its financial status. The reports built from that record are called accounts, and together they form the financial statement of the business.

This chapter explains the three main components of that statement - the balance sheet, the income statement (also called the profit and loss account), and the cash flow statement - and how to read them carefully. You will learn the balance-sheet identity, how assets and claims on assets are classified, how profit relates to equity, why depreciation matters, why profit is not the same as cash flow, how financial accounting differs from management accounting, and how common ratios help you judge liquidity, profitability, leverage and efficiency. Share-related figures that investors use when reading published statements are included where they support analysis.

6.1 What a balance sheet is

A balance sheet is a snapshot of what a business owns and how that ownership was financed on a particular date. On one side it lists assets - the resources controlled by the business and used in its operations. On the other side it lists claims on those assets: liabilities (amounts owed to outsiders) and owner's equity (the residual claim of the owners). Because every euro bound in an asset must have come from somewhere, the two sides always add up to the same total.

Balance sheet. A statement of a company's assets and of how those assets were financed by liabilities and owner's equity at a specific point in time.

Assets. Resources the business owns or controls that are used for the business and that have economic value (for example equipment, inventory, receivables and cash).

Liabilities. Debts and obligations owed to other persons, businesses or banks that must be repaid at a certain point in time and/or over a certain period. Funds are often provided by banks or suppliers; money owed to suppliers is also called trade credit or trade payables.

Owner's equity (capital). The residual interest in the assets after deducting liabilities - the portion of assets not financed by debt. It reflects funds provided by owners or shareholders and profits retained in the business.

Balance-sheet identity: $\text{Assets} = \text{liabilities} + \text{owner's equity}$

This identity must hold because (1) all assets were funded either through equity or through liabilities, (2) all funds are somehow bound or invested in the business - even cash held in a bank account was funded through equity or liabilities and is now bound in that form - and (3) any increase in assets must be financed by an increase in either liabilities or equity.

Assets	Liabilities + Equity
Non-current assets	Equity
Current assets	Non-current liabilities
Inventory	Current liabilities
Receivables	Trade payables
Cash	Short-term loans
<i>Assets = Liabilities + Equity</i>	

The balance sheet: assets financed by liabilities and owner's equity.

Worked example - constructing a balance sheet. A newly founded consultancy holds the following resources on 1 January: office equipment and computers used in the office €25,000; a delivery van €8,000; computers held in stock for resale (inventory) €12,500; cash and bank deposits €3,500. A bank loan of €25,000 financed part of these purchases; the rest was financed by the owners.

Total assets = 25,000 + 8,000 + 12,500 + 3,500 = €49,000.

Liabilities (bank loan) = €25,000.

Owner's equity = assets – liabilities = 49,000 – 25,000 = €24,000.

Notice that office computers (used in operations) and computers held for resale are different types of asset even though the physical items look similar: classification follows intended use, not appearance.

Assets	€	Liabilities and equity	€
Office equipment (incl. computers)	25,000	Owner's equity (capital)	24,000
Van	8,000	Bank loan (liabilities)	25,000
Inventory	12,500		
Cash and bank deposit	3,500		
Total assets	49,000	Total liabilities and equity	49,000

Simplified opening balance sheet (€)

Assets are usually listed in a standard order. Fixed assets (also called non-current assets) normally have a lifespan of more than one year and are intended to be used in the company for longer than one year; they usually cannot be turned into cash so easily. Current assets have higher liquidity: they are usually not used longer than a year because they are used up, spent in production, or sold. Examples of non-current assets include property, plant, premises, buildings, machinery, office equipment (including computers used in the office), and financial assets meant to be kept longer than a year. Examples of current assets include inventory (merchandise not yet sold), accounts receivable / debtors (trade receivables - money owed to the business), and cash. Intangible assets such as trademarks, patents, copyrights and acquired licences cannot be seen or touched but still have value; they are assets just as tangible items are.

Non-current (fixed) assets. Assets with a useful life of more than one year that are intended for longer-term use in the business.

Current assets. Assets with higher liquidity that are normally used up, sold or converted into cash within one year.

Liabilities are split by timing. Current liabilities are debts or obligations due within one year. Non-current liabilities have a duration of more than one year. Equity is the difference of assets and liabilities and is an indicator of the wealth of the company.

Equity ratio: $\text{Equity ratio} = \text{equity} / \text{total capital (total assets)}$

A high equity ratio indicates that a larger portion of assets was financed by the company's own resources (funds from owners or shareholders) rather than by creditors. Sufficient equity matters because (1) equity usually does not have to be repaid on a fixed schedule, (2) it helps the business stay relatively independent from its creditors, and (3) if the company records a loss, a larger equity cushion reduces the risk of over-indebtedness. Using the opening balance sheet above: equity €24,000, total assets €49,000, so equity ratio = $24,000 / 49,000 \approx 0.490$, or about 49%. Almost half of the assets were financed by own resources; the remainder (about 51%) was financed by the bank loan.

Changes in the balance sheet depend on how a purchase is financed. If a firm buys software and pays cash, one asset (software) rises and another (cash) falls by the same amount: total assets remain unchanged. This is an asset swap. If instead the firm buys the software on credit, assets rise and liabilities rise by the same amount, so the balance-sheet total increases. The identity still holds in both cases. Buying an asset with cash does not automatically increase total assets; total assets rise only when the purchase is financed by new liabilities or new equity. Valuing assets is important and tricky. The outstanding amount of a bank loan is usually clear, but the worth of a building many years after it was built is harder to pin down. Businesses may wish to present themselves favourably, yet they must follow legal rules and regulations so that balance-sheet information can be trusted. Carrying amounts are not the same as market prices.

Worked example - asset swap versus financed purchase. Start from total assets €49,000, liabilities €25,000, equity €24,000.

Case A - buy €2,000 of software for cash:

Software +2,000; cash -2,000. Total assets still €49,000. Liabilities and equity unchanged.

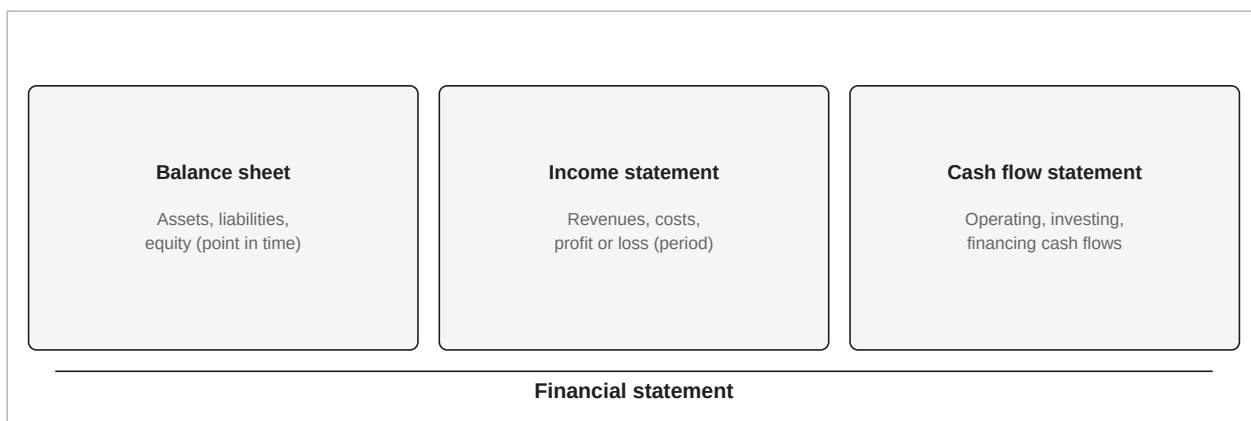
Case B - buy the same €2,000 of software on credit:

Software +2,000; trade payables +2,000. Total assets €51,000; liabilities €27,000; equity still €24,000.

Identity: $51,000 = 27,000 + 24,000$.

6.2 Other components of the financial statement

The balance sheet alone cannot show how a business performed over a period. It states assets, liabilities and equity on a certain day, but it does not report total sales (turnover), the cost of producing goods and services, or other expenses. The financial statement therefore combines three main components: the balance sheet (assets, liabilities and equity), the income statement or profit and loss account (revenues, costs and expenses), and the cash flow statement (inflows and outflows of cash). Together they reveal performance over time as well as the position at a date.



Main components of a financial statement: balance sheet, income statement and cash flow statement.

Income statement (profit and loss account). A period statement that summarises all revenues, costs and expenses over a time span (for example a year) to determine whether the business made a profit or a loss.

Revenues. Income (usually cash or accounts receivable) generated from selling goods or services to customers.

Costs. Resources consumed in order to produce the goods or provide the services sold.

If revenues exceed costs and expenses, the company has a profit. If costs and expenses are higher than revenue, it suffers a loss. Profit for the year increases owner's equity (typically through retained earnings); a loss decreases equity. You can therefore learn about profit either from the income statement or from the change in equity between two balance-sheet dates (other owner transactions aside).

Worked example - simple profit. In one financial year a service firm records sales revenue of €400,000 and costs and expenses (materials, wages, rent, energy, depreciation and other expenses) of €310,000.

$\text{Profit} = 400,000 - 310,000 = \text{€}90,000.$

That profit increases equity by €90,000 if it is retained in the business.

Cost of sales (also called cost of goods sold) covers costs directly tied to production - materials, labour linked to production, and manufacturing overhead. Administration costs, shipping to customers and sales-staff costs are not included in cost of sales; they are operating expenses below gross profit. Gross profit shows earnings after deducting the direct costs of producing the goods, before those further operating expenses. In accounting, expenditures and expenses are not the same. Expenditures are payments made to purchase assets (non-current or current). Expenses are costs that have expired or been "used up" in producing the goods or services sold - for example cost of sales, salaries, marketing, interest, insurance and rent.

Depreciation. Recognition that the value of fixed assets decreases as they are used up over time. Without depreciation, asset values in the accounts would be overstated and financial information flawed. Depreciation is an expense in the income statement, but unlike wages or energy it does not cause an actual cash payment in the period when it is charged.

Straight-line depreciation (per year): $\text{Annual depreciation} = \text{depreciable cost} / \text{expected useful life}$
(Equal charge each year; land is not depreciated.)

Worked example - straight-line depreciation. A firm buys a computer for €2,100 and expects to use it for three years (expected useful life = 3 years), with no residual value for simplicity.

$\text{Annual depreciation} = 2,100 / 3 = \text{€}700 \text{ per year.}$

Book (carrying) values at year-end:

After year 1: $2,100 - 700 = €1,400$.

After year 2: $1,400 - 700 = €700$.

After year 3: $700 - 700 = €0$.

Each year the income statement includes a €700 depreciation expense that reduces profit, while the cash outlay of €2,100 occurred when the computer was purchased (an investing cash outflow if paid then), not when depreciation is later recognised. Typical useful lives in practice vary by asset class (buildings longer than machinery; tools and office equipment often a few years). Land is not subject to depreciation.

Over a trading year, the balance sheet also changes for many other reasons: new equipment may be bought, inventory levels adjust, customers may still owe money (receivables), the firm may still owe suppliers (trade payables), and part of a bank loan may be repaid. Equity rises when the firm is profitable and falls when it makes a loss. The following year-end statement illustrates those movements after a profitable first year.

Assets	€	Liabilities and equity	€
Office equipment (incl. computers)	42,000	Owner's equity	114,000
Van	40,000	Bank loan	12,500
Car	6,000	Trade credit / payables	15,000
Inventory	18,000		
Receivables	12,500		
Cash and bank deposit	23,000		
Total assets	141,500	Total liabilities and equity	141,500

Simplified year-end balance sheet after a profitable year (€)

Cash flow statement. A statement that reveals the flows of cash into and out of the business during a period, distinguishing operating, investing and financing activities. Cash is needed for day-to-day operations and for investing; a positive cash flow is not identical with a profit.

Cash flow from operations shows how well a business generates cash with its core activities (for example collecting receivables, paying for materials, settling trade credit). This section is usually treated as the most important and should ideally be positive. Cash flow from investments shows cash spent on long-term assets (plant, office equipment) or received from selling such assets. A negative investing cash flow often simply means the firm invested; it is not automatically a crisis signal. Cash flow from financing refers to cash from investors or creditors, and outflows for interest in some presentations, dividends, or debt repayment. At the bottom of the cash flow statement, the net change in cash and cash equivalents reconciles opening and closing cash. If that figure is positive, more cash is available for further activities such as investment, expansion or debt repayment.

Worked example - total cash change versus profit. Opening cash (including bank deposits) is €3,500; closing cash is €23,000.

Total cash increase = $23,000 - 3,500 = €19,500$.

Suppose the same year reports a profit of €90,000. Profit and the cash increase differ because profit includes non-cash charges such as depreciation and accruals (sales on credit, unpaid expenses), while the cash figure only counts actual cash movements. A firm can be profitable yet short of cash, or can raise cash through borrowing without earning a matching profit.

Worked example - classifying cash flows. Classify these cash movements for a manufacturing firm in one year:

- Customers pay €120,000 of trade receivables -> operating inflow.

- Raw materials paid in cash €45,000 -> operating outflow.
- New machine bought for €80,000 cash -> investing outflow.
- Bank loan repayment €15,000 -> financing outflow.
- Dividend paid to owners €10,000 -> financing outflow.

Net operating cash (simplified here) = 120,000 – 45,000 = €75,000 inflow.

Investing cash = –€80,000. Financing cash = –15,000 – 10,000 = –€25,000.

Approximate net cash change from these items = 75,000 – 80,000 – 25,000 = –€30,000.

6.3 What can be learnt from reading a balance sheet and an income statement

Balance sheets and income statements should always be read with caution. Asset values depend on valuation rules; depreciation may lower carrying amounts below what an outsider thinks an asset is "really" worth. Still, careful reading - especially comparing positions over time and against competitors in the same industry - answers important questions about structure and performance.

From the balance sheet one asks: Which assets does the business have - is there a higher percentage of current or non-current assets, and is that mix typical for this type of business? How has equity developed, and how have the assets been financed? Is there a higher share of long-term or short-term liabilities? Is there a balance between non-current assets and long-term financial resources (equity plus non-current liabilities)? Long-term assets should preferably be financed with long-term funds. If non-current assets clearly exceed the sum of equity and non-current liabilities, part of the long-term asset base is implicitly financed by short-term liabilities - a structural risk when those short-term claims must be refinanced.

From the income statement one asks: How have revenues developed? Have costs - especially cost of sales - moved in line with revenues? How have profits or losses developed? A single year's profit or asset mix is less informative than trends over several years and comparisons with similar firms in the same sector. Industry norms for fixed-asset intensity differ sharply between manufacturers and service firms.

Worked example - reading structure and performance (euros in thousands). Suppose a manufacturing company reports at year-end: non-current assets 944; current assets 586; total assets 1,530. Equity 711; non-current liabilities 515; current liabilities 304; total equity and liabilities 1,530.

Share of non-current assets = $944 / 1,530 \approx 61.7\%$. A high fixed-asset share is typical of manufacturing that needs plant and machinery.

Equity ratio = $711 / 1,530 \approx 46.5\%$ - almost half of assets financed by equity.

Long-term finance available = equity 711 + non-current liabilities 515 = 1,226, which exceeds non-current assets 944. So long-term assets are covered by long-term resources.

Income statement (same year versus prior year): revenue rises from 815 to 992 (+177, about +21.7%), while cost of sales rises from 760 to 830 (+70, about +9.2%). Gross profit therefore rises sharply (from about 55 to about 162). Operating profit also improves strongly. Revenues grew faster than cost of sales - a favourable operating development when reading the income statement.

Two further earnings ideas appear when statements are read in detail. Earnings before interest, taxes, depreciation and amortisation add back non-cash charges to focus on operating cash-like performance; deducting depreciation and amortisation yields earnings before interest and taxes, a common operating-result measure. You do not need to memorise every line of a published statement, but you should know that gross profit focuses on production margins, while operating result also reflects distribution and administration costs.

6.4 Types of accounting - financial versus management accounting

Accounts exist because many stakeholders need reliable information on the financial situation of a business. Internal users include owners, managers and employees: they want to know whether the

business is thriving. Managers decide on cost cuts, pricing and investment; owners ask whether the return justifies the capital at risk. External users include tax authorities, suppliers, competitors, investors, banks and the media.

Financial accounting. Accounting aimed at producing information - such as the balance sheet and the income statement - that is of interest both inside the firm and to external decision makers (tax authorities, banks, investors). Key outputs appear in annual reports and, for many large businesses, in selected figures published on company websites.

Managerial (management) accounting. Accounting focused on providing information for the management of the business so managers can decide where to cut costs, how to calculate prices, and how to allocate resources. It is primarily an internal decision-support tool.

Auditing. Independent checking of accounts for authenticity by an auditing company. For some companies this check is mandatory. The result of the auditing process appears in the annual report.

Managers use both forms: they need management-accounting detail for operational decisions, and they also care about financial-accounting outcomes because banks, tax authorities and investors read those published figures. The same transaction may therefore feed both systems, but the reports differ in audience, detail, legal form and often in time horizon (historical external reporting versus forward-looking internal analysis). Managerial accounting does not exist mainly to satisfy tax authorities; it primarily supports internal decisions. Tax and creditor reporting belong to the domain of financial accounting (subject to local law). Internal users need accounting information as much as external users do.

Worked example - who needs which information. A bakery considers raising the price of a specialty bread. Management accounting estimates cost per loaf (€1.80 materials and labour), contribution after variable costs, and effect on monthly volume if the retail price rises from €3.20 to €3.50.

Separately, the bank financing a new oven asks for the latest balance sheet and income statement (financial accounting): total equity €90,000, bank loan €40,000, last year's profit €25,000. Tax authorities use the same financial accounts to assess taxable profit.

Both information sets matter, but they answer different questions for different users.

6.5 Analysis of financial statements

Figures from the balance sheet and income statement can be turned into ratios that shed light on liquidity (or solvency), profitability (profit relative to capital employed or turnover), financial efficiency (how effectively resources are used), and financial structure (equity ratio and debt ratio). Because ratios vary widely across industries, comparisons are meaningful mainly within the same industry or sector, or for one business over several years. There may also be different accepted ways to calculate a given ratio - always check the definition before comparing. Making an absolute profit does not automatically mean the business is sufficiently profitable: profitability ratios relate profit to equity, capital employed or turnover.

Theme	Ratio	Formula
Structure	Equity ratio	equity / total assets
Structure	Debt ratio	total debt / total assets
Liquidity	Working capital	current assets – current liabilities
Liquidity	Current ratio	current assets / current liabilities
Liquidity	Acid-test ratio	(current assets – inventory) / current liabilities
Profitability	Return on equity	profit before interest and tax / equity
Profitability	Return on capital employed	profit before interest and tax / capital employed
Profitability	Capital employed (one definition)	equity + non-current liabilities (or assets – current liabilities)
Efficiency	Asset turnover	turnover / average assets
Efficiency	Inventory (stock) turnover	cost of sales / average inventory
Shares	Earnings per share	profit for ordinary shareholders / shares outstanding
Shares	Price-earnings ratio	share price / earnings per share
Shares	Dividend yield	dividend per share / share price
Shares	Market capitalisation	shares outstanding × market price

Selected ratio formulas used in financial analysis

Liquidity is the ability of a business to pay its bills and repay its debts on time. One approach uses working capital (circulating capital), which focuses on short-term assets and liabilities and indicates whether the firm can cover day-to-day bills such as electricity, rent and wages and buy inputs for production. Working capital should be positive: current assets above current liabilities. If not, part of non-current assets may have been financed by short-term liabilities, which can cause trouble when those liabilities fall due. A low working capital level may also mean too little cash or too much reliance on trade credit. Retailers that hold a lot of inventory often need higher working capital than pure service businesses. High inventory helps meet customer demand quickly but ties up money, takes space, and risks obsolescence.

Working-capital and cash-flow problems are linked but not identical. Short-term borrowing or overdrafts can ease a cash shortage yet raise current liabilities and thus reduce working capital. Improving both often requires more equity or more long-term credit. The current ratio should exceed 1 and is often considered healthy roughly between 1.5 and 2 (rules of thumb, industry-dependent). The acid test excludes inventories for a stricter liquidity check, because inventories may not turn into cash reliably.

Worked example - liquidity ratios (euros in thousands). Current assets 586; inventory 136; current liabilities 304.

Working capital = $586 - 304 = 282$.

Current ratio = $586 / 304 \approx 1.93$ (above 1, near the common 1.5-2 range).

Acid-test ratio = $(586 - 136) / 304 = 450 / 304 \approx 1.48$ (still above 1).

Interpretation: short-term resources exceed short-term claims even after excluding inventory, so liquidity looks adequate on these measures - subject to industry comparison.

Profitability relates profit to the size of the business. A small absolute profit may be inadequate relative to capital invested and risk taken. Common measures use profit before interest and tax relative to equity or to capital employed. Capital employed is often equity plus non-current liabilities (or assets minus current liabilities). Average capital employed (average of opening and closing) is sometimes used instead of a

period-end figure.

Worked example - return ratios (euros in thousands). Operating profit before interest and tax 90; equity 711; total assets 1,530; current liabilities 304.

Return on equity = $90 / 711 \approx 12.7\%$.

Capital employed = $1,530 - 304 = 1,226$.

Return on capital employed = $90 / 1,226 \approx 7.3\%$.

If average capital employed were 1,168, return on capital employed = $90 / 1,168 \approx 7.7\%$.

These figures are most useful when compared with peers in the same industry and with the firm's own history. Investors typically prefer comparatively high returns on capital employed, other things equal.

Financial efficiency asks how well resources generate activity. Asset turnover relates turnover (revenue) to average assets or to average net assets (assets minus long-term liabilities). Inventory (stock) turnover relates cost of sales to average inventory and shows how many times stock was sold or used and replaced in a year. High inventory turnover usually means goods move quickly and less cash is tied up in stock.

Worked example - turnover ratios (euros in thousands). Turnover 992; assets beginning 1,437; assets ending 1,530.

Average assets = $(1,437 + 1,530) / 2 = 1,483.5$.

Asset turnover = $992 / 1,483.5 \approx 0.67$ (about €0.67 of sales per €1 of average assets).

Cost of sales 830; *average inventory* 122.5.

Inventory turnover = $830 / 122.5 \approx 6.8$ times per year -> stock renewed roughly every $365 / 6.8 \approx 54$ days.

When analysing listed companies, investors also use share-related figures published with financial results. These bridge the income statement and market prices. Earnings per share divide profit attributable to ordinary shareholders by the number of shares outstanding. A share buyback reduces shares outstanding and can raise earnings per share even if total profit is unchanged. The price-earnings ratio divides the current share price by earnings per share: a ratio of about 16 means an investor pays roughly €16 of price for €1 of earnings. A low price-earnings ratio may suggest undervaluation or simply high current earnings; a high or rising ratio may mean the shares look expensive relative to earnings, or that investors expect stronger future growth. Dividend yield expresses the dividend relative to the share price. Market capitalisation (shares outstanding $\times$ market price) gauges size but is not automatically a measure of fundamental value, because prices move for many reasons. After shares already trade on an exchange, a rise in market price does not by itself bring new cash into the company - only sellers benefit; new equity cash comes from issuing new shares.

Worked example - earnings per share and price-earnings ratio. Profit attributable to ordinary shareholders €5,000,000; shares outstanding 2,000,000; market price €20.

Earnings per share = $5,000,000 / 2,000,000 = €2.50$.

Price-earnings ratio = $20 / 2.50 = 8$.

An investor pays €8 of price per €1 of current earnings.

If the firm buys back 200,000 shares and profit stays €5,000,000, shares outstanding become 1,800,000 and earnings per share rise to $5,000,000 / 1,800,000 \approx €2.78$ - higher per-share earnings without any change in total profit.

End of the Full Course theory chapters

Return to the Economics Full Course practice area for the chapter you have studied. Read the figures and tables carefully — they summarise relationships that often appear in exam statements.